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Explainable AI Tutor for Strategic Decision-Making in *Catan*

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Abstract

As AI systems are increasingly used not only to play games but to support human learning, a central challenge is balancing strategic performance with explainability. This challenge is particularly relevant in the board game *Catan*, where stochastic outcomes, hidden information and multi-player interactions make strategic decision-making difficult to communicate in human terms. While many game-playing agents achieve strong performance through search or learned policies, their decision processes are often difficult to explain.

This project presents an explainable AI tutor for *Catan* that combines weighted heuristic evaluation and planning, evolutionary optimisation and explanation generation within a unified framework. A complete simulation environment was implemented to support gameplay, state evaluation, strategic planning and interactive tutoring. The resulting system is capable of providing move recommendations, strategic feedback and natural-language explanations during play.

Evaluation combined large-scale gameplay experiments with a small user study. Results indicate that the tutor achieved competitive gameplay performance while providing recommendations and explanations that users found useful for strategic decision-making and reflection. Overall, the project demonstrates how interpretable AI can support strategic tutoring in complex stochastic games.

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Chapter 1

Introduction

Strategic board games provide well-defined experimental domains for artificial intelligence research. Classic games such as *Chess* and *Go* have driven major advances in search algorithms and learning systems, resulting in highly capable deep learning models [1]. However, many popular board games introduce forms of complexity, including stochastic outcomes, hidden information, interactions and negotiation, which challenge both traditional search methods and contemporary data-hungry learning systems. These domains demand decision-making approaches that can operate effectively with limited data, adapt to changing strategic contexts and remain comprehensible to human players.

In this project, the focus is on the board game *Catan* by *Klaus Teuber*. *Catan* presents a combination of properties that make it substantially more challenging than deterministic, two-player games [2]. Dice rolls introduce randomness. Trading and negotiation permit significant cross-player influence. The branching factor is extremely large and imperfect information restricts what can be inferred from opponents' actions. Unlike many well-studied game domains that provide large-scale gameplay datasets, *Catan* lacks widely available public training resources and presents a state space far more irregular than grid-based environments [3]. As a result, many existing approaches rely heavily on hand-crafted heuristics, search methods or reinforcement learning techniques [4, 5].

Explainability becomes particularly valuable when AI systems are used as tutors or decision-support tools rather than purely competitive agents. Whereas a high-performing agent may be sufficient for autonomous gameplay, a tutoring system must justify its decisions in human-interpretable terms. Players benefit not only from being told what to do, but from understanding why certain actions are strategically favourable. Many high-performing neural approaches rely on complex internal representations that are difficult to interpret directly, while rule-based agents often depend heavily on domain-specific heuristics and careful tuning to remain strategically competitive [6]. This creates an open research challenge: designing a *Catan* agent that achieves strong strategic performance while also producing explanations that support human learning.

This project therefore investigates whether interpretable rule-based agents can remain strategically competitive in a complex stochastic game such as *Catan*, whether evolutionary optimisation can improve gameplay performance without sacrificing transparency and whether decision-level explanations can support useful tutor-style feedback for human players. More broadly, the project explores the trade-off between strategic strength, explainability and educational usefulness under uncertainty and player interaction.

1.1 Definition of AI in this Work

Throughout this work, the term *AI agent* is used in a broad sense to refer to automated decision-making systems capable of evaluating game states and selecting actions. This includes rule-based systems, search-based methods, reinforcement learning agents and evolutionary optimisation approaches. While some discussions reserve the term "AI" primarily for machine learning or large-scale neural models, game AI research traditionally encompasses both hand-crafted heuristic agents and learning-based approaches. This broader definition is adopted here, as the proposed tutor combines interpretable rule-based reasoning with optimisation and simulation-based evaluation.

1.2 Objectives and Contributions

In particular, this project sets out to accomplish the following objectives:

- Develop an explainable decision framework for *Catan* based on the Expected Time to Win planning approach, combining weighted rule-based reasoning with strategic planning while balancing gameplay performance and human interpretability.
- Automatically optimise the agent's strategic evaluation parameters through evolutionary algorithms, improving gameplay performance while preserving the transparency of the underlying decision process.
- Implement a complete simulation environment to support experimentation and evaluation, including a detailed game state representation, feature extraction pipeline and simulation-based agent evaluation framework.
- Produce a prototype capable of generating tutor-style explanations grounded directly in the agent's planning and utility evaluation process, demonstrating how the system can provide human-interpretable strategic guidance.
- Systematically evaluate the proposed approach using both quantitative performance metrics and qualitative human-centred criteria, investigating how users interact with tutor recommendations and explanations and assessing their perceived usefulness, interpretability and educational value.

The intended contributions of this project include the design and implementation of a fully explainable *Catan* agent with automatically optimised behaviour, alongside a reusable simulation and evaluation framework for experimentation. Collectively, this work seeks to establish a foundation for explainable tutoring and decision-making in complex, imperfect-information games.

Chapter 2

Background and Related Work

2.1 Rules of *Catan*

Catan is a multiplayer strategy board game in which 4 players compete to reach 10 victory points. Players accumulate points by constructing settlements, upgrading settlements to cities and achieving special awards or using development cards. The game combines stochastic events, hidden information, negotiation and strategic planning, making it a complex environment for both human and AI players.

The standard rules and detailed descriptions of game components can be found in the official *Catan* game manual [7]. An example of the board layout is shown in Figure 2.1.

Figure 2.1: *Catan* game board showing roads and settlements. Source: [7].

2.1.1 Game Board and Setup

The board consists of a hexagonal grid of land tiles representing five types of resources: wood, brick, sheep, wheat and ore, along with a desert tile that produces no resources. Each land tile is assigned a number token from 2 to 12 (excluding the desert), representing the dice roll required for it to produce resources. Around the board, ports allow players to trade resources with the bank at favourable rates.

Each player begins by placing two settlements and two connecting roads on intersections and edges of hexes during an initial placement phase. Placement strategy is crucial: it affects access to resources, proximity to ports and future expansion potential.

2.1.2 Resource Collection

At the start of each turn, the active player rolls two six-sided dice. Each land tile with a number matching the dice roll produces one resource of that type for every adjacent settlement, or two resources for a city. Resource production is stochastic, introducing variability in available actions.

2.1.3 Construction and Development

Players can spend resources to perform several core actions during their turn:

- **Build Roads**: Expand a player's network, enabling new settlement placements and increasing access to key resources.
- **Build Settlements**: Provide one victory point and access to resource production from adjacent tiles.
- **Upgrade to Cities**: Replace settlements, doubling resource production from adjacent tiles and increasing victory points.
- **Buy Development Cards**: Draw a hidden development card that may provide strategic advantages, including:
  - **Knight cards**: Move the robber and steal one resource from an opponent.
  - **Road Building cards**: Place two free roads.
  - **Year of Plenty**: Gain resources directly from the bank.
  - **Monopoly cards**: Claim all resources of a specific type from other players.
  - **Victory Point cards**: Provide hidden victory points revealed only when the game ends.

2.1.4 Special Mechanics and Achievements

- **The Robber**: Activated when a player rolls a 7. Players with more than seven resources must discard half. The robber moves to a new hex, blocking its resource production and allowing the active player to steal one resource from an adjacent opponent.
- **Longest Road and Largest Army**: These achievements each award two victory points. Longest Road requires at least five connected road segments. Largest Army requires playing at least three knight cards.

2.1.5 Trading and Negotiation

Players may trade resources with each other or use ports to exchange with the bank. Trade introduces a large branching factor and adds significant complexity. Negotiation allows players to propose trades, accept or reject offers and counteroffer. Human players can execute nuanced strategies, including deception, persuasion and coalitions.

2.2 Complexity and Challenges for Game AI Agents

Catan is a partially observable, stochastic environment:

- Extremely large state space due to variable board layouts, dice outcomes and trading interactions ($\approx 1.2 \times 10^{15}$ initial states) [3].

- Hidden information includes opponents' resource cards, unplayed development cards and intended future actions, meaning agents cannot fully observe the strategic state of other players.
- Stochastic events such as dice rolls.
- Strategic interactions between multiple players, where each agent acts independently with competing objectives in a shared game environment.
- Diverse action types: building, trading, card usage, negotiation, ending turn.

Success requires balancing short-term tactical decisions (e.g., trading or stealing resources) with long-term strategic planning (e.g., aiming for Longest Road or Largest Army). Agents can exploit structured regularities, such as predictable probabilities and common high-level strategies, while adapting to opponent behaviour.

2.3 AI Approaches for *Catan*

Games, such as *Catan*, provide a structured yet uncertain decision-making environment, combining clear rules, randomness and strategic interactions between players. As a result, there are various AI techniques that have been explored when dealing with planning, learning and evaluation, under uncertainty. This section reviews the key approaches previously applied to *Catan*, focusing on their effectiveness and explainability.

Many studies evaluate learning-based agents against established rule-based systems, most notably the JSettlers framework, a heuristic-driven *Catan* AI derived from earlier rule-based research by Thomas [8]. While this section focuses on search and learning approaches, rule-based strategies are discussed in detail in Section 2.4.

2.3.1 Monte Carlo Tree Search

Method Overview

Monte Carlo Tree Search (MCTS) is a search algorithm that has been widely adopted for games with large branching factors and inherent uncertainty. Rather than exhaustively enumerating all possible game states, MCTS incrementally builds a search tree through repeated simulations. It works by balancing exploration of less-visited actions with exploitation of actions that have produced favourable outcomes [2]. In this work, the MCTS agent evaluated by Szita et al. is referred to as Szita-MCTS for clarity.

Each MCTS iteration consists of four phases:

1. Selection: Traversing the tree from the root by selecting child nodes according to a policy such as the Upper Confidence Bound.
2. Expansion: Adding one or more child nodes when a non-terminal leaf node is reached.
3. Simulation (Rollout): Executing a random or policy-guided simulation until a terminal state is encountered.
4. Backpropagation: Propagating the simulation result back through the visited nodes to update their value estimates.

MCTS works well for *Catan*, because it handles stochastic elements (e.g. dice rolls), uncertain opponent behaviour and a large action space. It samples possible future game trajectories, enabling it to estimate the long-term utility of actions, such as early road expansion, settlement placement or resource-focused strategies. Empirical studies show that the performance results of MCTS, with sufficient simulation depth, outperform purely rule-based agents.

Empirical Performance

In the SmartSettlers environment, Szita-MCTS-10,000 achieves a 49% win rate against three rule-based JSettlers agents over 100 games [2]. In contrast, a Random agent achieves a 0% win rate against the same opponents. This suggests that Random play may be too weak to provide a meaningful baseline in this setting.

Interpretability Limitations

Despite its strong performance, MCTS has limited interpretability because action choices emerge from aggregated simulation outcomes rather than explicit strategic reasoning. For example, an MCTS agent may recommend building a road because rollouts beginning with that action achieved higher average rewards across thousands of simulated futures. However, the search process does not naturally expose a concise explanation of *why* the road is strategically valuable, such as whether it improves expansion opportunities, resource access or future settlement potential. Explanations are therefore often limited to describing simulation success rather than communicating the goal-oriented reasoning expected from a tutoring system.

2.3.2 Reinforcement Learning

Method Overview

Reinforcement Learning (RL) addresses the problem of learning how to act optimally through trial-and-error interaction with an external environment [4]. In this work, the reinforcement learning agent proposed by Gendre and Kaneko is referred to as Gendre-RL for clarity. At each discrete time step t, the agent is situated in a state s_t and selects an action a_t. As a consequence, it receives a scalar reward r_t and the environment transitions to a subsequent state s_{t+1}.

The agent's behaviour is characterised by a policy $\pi(a \mid s)$, which specifies how actions are selected given states. Learning in RL aims to identify a policy that maximises the expected long-term return, defined as the discounted sum of future rewards:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k},$$

where the discount factor $\gamma \in [0, 1]$ controls the relative importance of immediate versus delayed rewards.

RL has been applied to *Catan*, enabling agents to learn strategies for settlement placement, road building and resource management. By interacting with the environment, RL agents can discover strategies that are difficult to specify directly from the rules alone. In principle, RL is well suited to stochastic environments and long-horizon decision making, though learning robust behaviour in the full multiplayer version of *Catan* remains challenging. In large state spaces, effective exploration is itself a major challenge for RL systems. Methods such as randomised value functions have been proposed to improve exploration efficiency by explicitly representing uncertainty in value estimates and encouraging more coherent exploration behaviour [9].

Empirical Performance

Recent work suggests that RL can achieve competitive performance in simplified *Catan* settings. Gendre and Kaneko report that Gendre-RL outperforms the rule-based JSettlers baseline in a two-player version of the game, achieving a win rate of approximately 56.5% after extended training [4]. This demonstrates that modern deep RL methods can learn strategically useful behaviour in constrained *Catan*-like environments.

A more recent implementation-focused study by Daijavad et al. also reports an RL-based *Catan* agent, though its results are less consistent across evaluation settings, suggesting that RL performance remains sensitive to training setup and opponent configuration [5].

However, these results should be interpreted with care. Much of the stronger reported RL performance in *Catan* comes from simplified settings, for example, reducing the game to two players or removing trading altogether [4]. Earlier work by Pfeiffer also found that learning a full high-level strategy directly was difficult and that better performance was obtained by combining learned low-level behaviours with rule-based heuristics [10]. Taken together, these studies suggest that while RL can learn useful strategic behaviour in constrained versions of *Catan*, robust performance in the full multiplayer game remains difficult because of the large action space, stochasticity and strategic interaction between players.

Interpretability Limitations

Like MCTS, RL is generally opaque, producing decision policies that are difficult for humans to interpret directly. For example, an RL agent may recommend building a city because its learned policy associates similar game states with higher expected long-term rewards. However, the decision process is typically encoded within learned value estimates or neural network parameters, making it difficult to identify which strategic factors contributed most strongly to the recommendation. Although RL can achieve strong strategic performance, it does not naturally expose the goal-oriented explanations that are desirable in tutoring systems. Recent work has begun to explore methods for making RL behaviour more understandable (see Section 2.5.3).

2.4 Rule-Based Strategies for *Catan*

Rule-based approaches provide a natural starting point for *Catan* AI because they encode strategic knowledge through explicit heuristics rather than learned black-box policies. In *Catan*, this is especially appealing for explainability: decisions can be justified in terms of interpretable concepts such as expected resource production, expansion potential, opponent threat and progress towards concrete goals. Early work by Thomas showed that a *Catan* agent could be designed around explicit planning heuristics rather than purely reactive play [8], while later work by Guhe and Lascarides demonstrated how these symbolic strategies could be refined and improved through simulation and comparison with human play [6].

A useful way to understand rule-based *Catan* agents is not as one fixed algorithm, but as a family of strategies built around the same idea: evaluating actions according to how much they improve a player's long-term path to victory. In practice, this often means reasoning in terms of sub-goals such as building a settlement, upgrading to a city, reaching a port, or progressing towards Longest Road or Largest Army.

2.4.1 Long-Term Planning through ETW and ETB

A central idea in Thomas's work is to represent strategy as a dynamic plan made up of intermediate goals, each of which has an associated *Estimated Time to Build (ETB)* [8]. At a high level, the overall *Estimated Time to Win (ETW)* can be viewed as the cumulative time required to complete successive milestones:

$$T_{\text{total}} = \sum_{i=1}^{n} T_i \tag{2.1}$$

where each T_i denotes the expected time required for a sub-goal in the current plan.

This allows actions to be compared according to how much they reduce the projected time to victory:

$$\Delta T_a = T_{\text{total(before)}} - T_{\text{total(after)}} \tag{2.2}$$

This framing is important because it links short-term tactical choices to long-term strategic outcomes. For example, building a road may not immediately increase victory points, but may still be optimal if it significantly improves access to a future settlement. Likewise, choosing to trade, wait, or buy a development card can be justified if it accelerates the overall route to a later goal. Since actions are evaluated in relation to explicit goals, this also makes the agent's reasoning inherently interpretable.

Guhe and Lascarides build on this general style of planning, but show that rigid priority orderings over build plans can be limiting [6]. Their work instead favours more flexible plan selection based on the current board state and available opportunities, rather than relying on fixed build priorities.

2.4.2 Resource Estimation and Build Planning

To reason about ETB, the agent must estimate how quickly resources can be acquired. Since *Catan* is driven by stochastic dice outcomes, rule-based agents typically use expected production rates derived from the probabilities of adjacent number tokens. For a resource type r, a player's effective production frequency may be approximated by

$$f_r = \sum_{h \in H_r} P(\text{number}(h)) \times \text{production\_factor}(h) \tag{2.3}$$

where H_r is the set of adjacent producing hexes and the production factor accounts for the increased resource output of cities relative to settlements

These frequencies can then be used to estimate how long it will take to collect a target set of resources. In practice, ETW-driven systems typically approximate expected resource acquisition using production probabilities and simplified trading assumptions, rather than modelling the full stochastic game process exactly. This allows candidate plans to be evaluated efficiently while still providing useful strategic comparisons.

Build planning also depends on distinguishing between theoretically valid and immediately reachable settlement locations. A settlement site may satisfy the distance rule but remain unavailable until it is connected to the player's road network. Consequently, planning must account not only for the final objective, but also for the intermediate infrastructure required to reach it. Trading and port access can be incorporated naturally into this framework, as they may reduce the ETB of future actions by improving access to required resources.

2.4.3 Utility-Based Action Selection

Once build times and candidate plans have been estimated, ETW-driven agents typically combine several strategic considerations into a single utility function. Thomas's work primarily emphasises self-progress through reductions in ETW, but later approaches also incorporate interference with opponents and progress towards secondary objectives such as Longest Road or Largest Army [6, 8].

A generic weighted formulation is:

$$U_{\text{total}}(a) = w_{\text{self}} \cdot U_{\text{self}}(a) + w_{\text{opp}} \cdot U_{\text{opp}}(a) + w_{\text{spec}} \cdot U_{\text{special}}(a) \tag{2.4}$$

where w_{self}, w_{opp} and w_{spec} control the relative importance of self-progress, opponent interference and special-objective progression, respectively.

The self-progress term is often defined directly in relation to ETW:

$$U_{\text{self}}(a) = \frac{\text{ETW}_{\text{before}} - \text{ETW}_{\text{after}}(a)}{\text{ETW}_{\text{before}}} \times 100 \tag{2.5}$$

This captures the intuition that an action is strong if it significantly shortens the player's expected path to victory. Opponent interference then adds value to moves that slow down dangerous opponents, while the special-objective term captures progress towards long-term strategic goals. A time discount can also be introduced so that delayed benefits are penalised relative to faster payoffs:

$$EU(a) = \frac{U_{\text{total}}(a)}{(1+d)^{\text{ETB}(a)}} \tag{2.6}$$

where d is a discount parameter.

A similar perspective appears in other *Catan* AI work. Pfeiffer describes agents in terms of high-level strategic modes such as expansion, city conversion, development-card acquisition and interference with opponents [10]. Although that work is not purely rule-based, it reinforces the idea that *Catan* strategy is naturally expressible in terms of explicit sub-goals and interpretable strategic components.

This kind of utility-based comparison is appealing because it allows different actions to be evaluated within one framework. A city, a road, a trade, or a development card purchase can all be assessed according to their contribution to future progress. Importantly, this same decomposition also supports explainability, since the selected action can later be justified in terms of the factors that contributed most strongly to its score.

2.4.4 Initial Placement and Early Positioning

Initial settlement placement is particularly important in *Catan*, since it determines early resource access, diversity and future expansion opportunities. Rule-based agents therefore often use a dedicated placement heuristic rather than relying on the same evaluation used during the main game.

A typical settlement utility balances strong production numbers, resource diversity and strategic positional value:

$$U_{\text{settle}}(s) = w_{\text{prod}} \cdot \sum_{h \in H(s)} P(\text{number}(h)) + w_{\text{div}} \cdot \text{ResourceDiversity}(s) - w_{\text{block}} \cdot \text{BlockingPenalty}(s) \tag{2.7}$$

where $H(s)$ denotes adjacent hexes, *ResourceDiversity* rewards access to multiple resource types, *BlockingPenalty* captures the cost of poor future expansion or strategic vulnerability and w_{prod}, w_{div} and w_{block} are tunable weighting coefficients.

Mathematical analyses of *Catan* opening play reach similar conclusions. Austin et al. show that settlement quality depends not only on expected production, but also on factors such as number diversity, resource scarcity and broader board context [11]. This suggests that effective placement heuristics should consider multiple strategic factors rather than raw probability alone.

Later work further refined these ideas. Guhe and Lascarides report that stronger opening heuristics, including discouraging overly redundant number coverage across the two initial settlements, improved performance relative to earlier symbolic agents [6]. The second settlement is therefore usually chosen with the first in mind, so that the pair provides a balanced starting portfolio rather than two individually strong locations that provide overly similar resource access.

2.4.5 Opponent Modelling, Trading and Robber Play

Unlike deterministic two-player games, *Catan* requires reasoning about multiple opponents whose actions affect the value of almost every decision. Rule-based agents therefore often include lightweight opponent modelling, estimating which opponent is currently strongest and how particular moves may change their projected progress.

Trading is especially important because it introduces both cooperation and competition. In Thomas's framework, a trade is not treated as beneficial simply because it yields a desired resource; it must also be better than available alternatives, such as waiting for production or using bank and port trades [8]. This can be expressed through the idea of a *Best Alternative To a Negotiated Agreement (BATNA)*: a proposed trade is worthwhile only if it improves the agent's expected progress more than its best non-agreement option. This makes trading naturally compatible with ETW-style planning, since the value of a trade depends on whether it accelerates the player's broader plan more effectively than its alternatives.

Robber placement and theft can be handled in a similar way, using heuristics that prioritise blocking high-probability resource tiles belonging to strong opponents. While these policies are relatively simple, they are effective because they translate highly interactive parts of *Catan* into explicit and explainable rules. Overall, ETW-driven rule-based systems are attractive not only because they can play competently, but because they provide a transparent foundation for tutor-style explanation.

2.5 Explainability in Game AI

Explainable AI (XAI) refers to AI agents whose decision-making can be understood, interpreted and communicated to humans. Games are ideal testbeds for XAI, since they have well-defined states, discrete actions and measurable outcomes.

2.5.1 Transparency of Agent Architectures

Rule-Based Systems In rule-based systems, behaviour comes from explicit heuristics, rules and structured evaluation functions. This means that each decision is traceable to a specific rule. For example, prioritising high-value resources or securing critical positions. This makes the thought process immediately interpretable [12].

Black-Box Learning Systems High-capacity statistical models (e.g., deep RL) achieve strong performance at the cost of low transparency [4]. These systems use internal representations such as neural activations and large policy tables, which can produce strong performance but are difficult to interpret, as they do not match the human thinking process.

Hybrid Approaches Hybrid agents combine learning with semantic representations [13]. For example, weights can be learned for predefined features, or policies can be constrained to interpretable action types. This keeps the flexibility of learning-based approaches and produces behaviour that humans can understand.

2.5.2 Explanation Systems in Deterministic Games

Deterministic games provide clear examples of practical XAI implementations. Platforms like Chess.com [14] and Lichess demonstrate that even high-performing engines whose internal evaluation processes are not directly human-readable can generate interpretable guidance. Typically, the engine evaluates possible moves, while a separate explanation module translates this evaluation into human-understandable terms, such as "best move", "inaccuracy", or common tactical patterns.

Recent research analysing neural-network-based chess agents, such as AlphaZero, has also explored ways to interpret deep models. Hammersborg and Strümke [15] propose an information-based approach using concept detection and trainable masking modules, which selectively highlight the parts of a board state that most influence the agent's decision. This allows humans to visualise and understand which pieces or positions are critical, demonstrating that even high-performing neural-network policies with complex internal representations can provide interpretable guidance when explanation systems are carefully designed.

These examples illustrate that interpretability can be achieved either through inherently transparent decision-making or through post-hoc explanation layers applied to complex models.

2.5.3 Explanation Systems in Uncertain Environments

Stochastic games introduce uncertainty through chance events such as dice rolls or card draws. Explaining agent decisions in these contexts requires accounting for probabilistic outcomes.

One approach for explaining decisions under uncertainty is to decompose the agent's evaluation into semantically meaningful components, such as trading potential, building opportunities, or resource accumulation. By comparing these components across possible actions, the agent can highlight which objectives drive its choices. Juozapaitis [13] introduces methods like Reward Difference Explanations and Minimal Sufficient Explanations, which show how agents can explain trade-offs between competing objectives in reinforcement learning environments. This provides interpretable insights into agent behaviour without requiring the user to understand complex internal computations.

Although these methods have been applied in environments such as *CliffWorld* and *Lunar Lander*, no equivalent XAI techniques currently exist for stochastic board games like *Catan*, leaving a clear research gap.

2.6 Tutoring Systems and Educational Design

AI agents used for learning differ from purely competitive systems in that their objective is not only to act effectively, but to support human understanding. Intelligent Tutoring Systems emphasise adaptive guidance, feedback and structured explanations that help learners reason about decisions rather than merely reproducing optimal behaviour [16].

2.6.1 Roles of a Game AI Tutor

Tutoring agents in games typically adopt one or more interaction roles. A *coach* provides guidance during gameplay, suggesting actions or drawing attention to risks in the current state. A *post-game analyst* explains decisions retrospectively, enabling reflection once outcomes are known. A *demonstrator* produces example gameplay that learners can observe and compare against their own choices. These roles impose different explanation requirements: in-game coaching benefits from concise, actionable reasoning, while analysis and demonstration allow more detailed comparison [17].

In this project, the primary role is an in-game coach. The tutor provides reasoning when relevant, while still allowing players to retain control over decisions. Demonstration and analysis behaviours are secondary and mainly support explanation rather than replace player agency.

2.6.2 Guidance and Scaffolding

A consistent finding in educational game research is that effective tutors provide *adaptive scaffolding* rather than constant instruction. Guidance should be responsive to context, offering support when learners encounter complex decisions without excessive intervention that may reduce engagement or independent reasoning [18]. In practice, this motivates selective explanation strategies: the tutor may provide advice on demand or highlight high-impact decisions, instead of continuously directing play.

Adaptive scaffolding also influences how explanations are phrased. Rather than presenting a single "correct" move, effective tutoring emphasises reasoning in terms of goals, constraints and trade-offs. This supports transferable understanding by connecting advice to strategic concepts learners can apply in future situations [19].

2.6.3 Implications for a *Catan* Tutor

Designing a tutor for a stochastic, multi-agent board game introduces additional considerations. Advice must account for uncertainty and opponent interaction, which means explanations should reference expected outcomes and strategic tendencies rather than deterministic guarantees. Educational perspectives on explainable AI emphasise that transparent reasoning can improve trust and engagement, particularly when explanations align with human-understandable concepts [20].

These principles inform the design of this tutor, which explains the strategic motivations behind recommendations while leaving the final decision to the player. By grounding explanations in interpretable features and structured reasoning, the system aims to support human learning while still encouraging independent decision-making.

This is particularly relevant for rule-based systems. Although decision-tree-style policies can in principle be exposed directly to users, Xu argues that such representations are often not especially useful in practice for human understanding [12]. This highlights the distinction between *interpretability* and *explainability*: a system may expose its internal reasoning structure, yet still fail to communicate its decisions in a form that is meaningful to users [21]. Instead, users tend to prefer dialogue-based explanations, such as asking why a recommendation was made or why an alternative was not chosen. For a tutoring system, explainability therefore depends not only on whether reasoning is transparent in principle, but also on whether it can be communicated in an accessible, interactive form [22].

2.7 Strategy Weights and Optimisation Techniques

In complex strategy games like *Catan*, an agent's behaviour can often be characterised as a combination of multiple strategic tendencies, such as prioritising trading, building infrastructure, or accumulating resources. These tendencies can be represented as weights within an evaluation function that guides decision-making:

$$V(s) = \sum_{i=1}^{n} w_i f_i(s), \tag{2.8}$$

where $f_i(s)$ are features of the game state (e.g., resource counts, board control, potential trades) and w_i encode the relative importance of each feature. Optimising these weights can significantly improve performance compared to manually tuned heuristics. Rather than redesigning the agent's decision rules, optimisation focuses on refining how strongly different strategic factors influence action selection, preserving interpretability while improving effectiveness.

Supervised Learning

Weights can be learned from datasets of expert moves or game outcomes using supervised learning [23]. The agent predicts the optimal action or expected outcome and adjusts its weights to minimise prediction error. This approach is effective at learning from existing strategies and can generalise when sufficient and diverse training data are available.

However, supervised learning requires labelled examples that represent desirable behaviour. In complex multi-agent environments such as *Catan*, large-scale public gameplay datasets remain limited and lack widely adopted standardised representations. Moreover, optimal actions are often context-dependent, influenced by hidden information, negotiation dynamics and stochastic events. As a result, optimisation methods that rely solely on prediction accuracy may struggle to capture long-term strategic value. These limitations motivate alternative approaches that optimise behaviour directly through interaction with the environment.

Evolutionary Algorithms

Evolutionary algorithms provide an optimisation framework that does not rely on labelled training data [24]. Candidate weight vectors are evaluated directly through gameplay simulation, with selection and mutation used to iteratively improve a population of strategies. Fitness is derived from observed performance outcomes rather than prediction targets, making evolutionary optimisation suitable for stochastic multi-agent environments.

Evolutionary optimisation can be viewed as an iterative loop around game simulation. Candidate solutions are initialised, evaluated through simulated matches and used to update the search process until a stopping criterion is reached. Figure 2.2 illustrates this optimisation cycle.

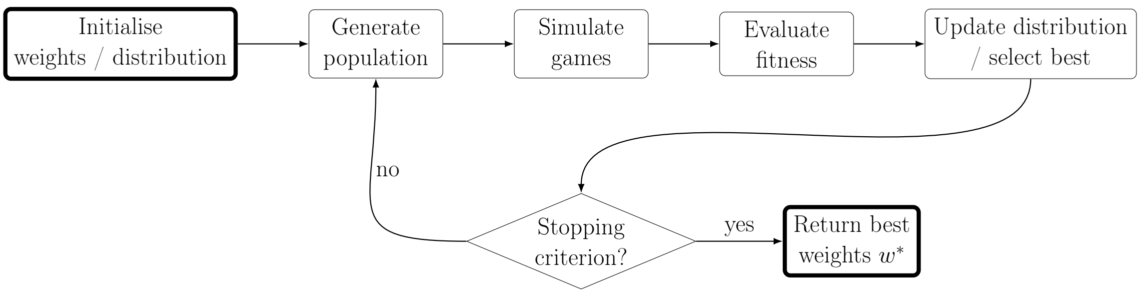


Figure 2.2: Evolutionary optimisation of strategy weights via simulation-based evaluation.

Evolutionary methods have been widely applied in game AI to tune heuristic evaluation functions and behavioural parameters [25]. Two broad paradigms are commonly distinguished. Genetic algorithms often operate on discrete encodings and emphasise crossover between solutions, whereas evolution strategies focus on continuous parameter optimisation through probabilistic mutation and adaptation. Since the strategy weights in this project are continuous, an evolution-strategy approach provides a natural optimisation framework.

Covariance Matrix Adaptation Evolution Strategy

Among evolution strategies, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is a widely used method for continuous black-box optimisation, where optimisation depends only on observed performance outcomes [26]. Candidate solutions are sampled from a multivariate Gaussian distribution whose covariance matrix is adapted over time. This enables the search process to capture correlations between parameters and progressively bias sampling towards promising regions of the parameter space.

CMA-ES is particularly suitable for tuning weighted rule-based agents. It does not require gradient information, allowing optimisation when performance is measured through simulation. Additionally, covariance adaptation supports coordinated adjustment of multiple strategy weights, encouraging balanced behaviours that are difficult to obtain through independent parameter mutations or manual tuning.

Because optimisation modifies only the numerical weights of the evaluation function, the underlying rule structure remains unchanged. This allows performance to improve while preserving the interpretability of the agent's decision-making.

2.8 Summary and Research Gap

This chapter has reviewed the rules and strategic complexity of *Catan*, highlighting the challenges of partial observability, stochastic events and hidden information. The three main AI approaches that have previously been implemented were reviewed and compared. Rule-based systems provide clear and interpretable decisions, while lacking adaptive depth. MCTS and RL methods learn more sophisticated strategies, but are largely opaque unless combined with still experimental techniques for explainability.

Empirical studies provide a snapshot of how these approaches perform in practice. Szita-MCTS-10,000 achieves strong performance against rule-based JSettlers agents in the standard multiplayer setting, while Gendre-RL achieves competitive performance in a simplified two-player setting without trading [2, 4]. Although these results are not directly comparable across evaluation settings, they illustrate the range of approaches explored for *Catan*, as summarised in Table 2.1:

Agent	**Setting**	**vs Rule-based (JSettlers)**
Random	4-player – Standard	0%
Gendre-RL	2-player – No Trading	56.5%
Szita-MCTS-1,000	4-player – Standard	27%
Szita-MCTS-10,000	4-player – Standard	49%

Table 2.1: Reported performance of different agents against rule-based JSettlers baselines.

Research in explainable AI has primarily focused on deterministic games such as chess or on simulated reinforcement-learning environments [13, 15]. Existing approaches typically separate decision-making from explanation by applying post-hoc interpretation techniques to complex policies. However, equivalent approaches have not been widely explored in socially interactive game environments such as *Catan*, where uncertainty, negotiation and long-term strategic planning make explanation substantially more difficult.

More broadly, prior work has largely treated gameplay performance, explainability and tutoring as separate problems. Existing *Catan* agents primarily focus on competitive play, while explainable AI research has concentrated on either deterministic games or simplified RL environments. Similarly, intelligent tutoring systems have rarely been explored in the context of complex stochastic board games. This creates a clear research gap at the intersection of game AI, explainable AI and intelligent tutoring systems.

This gap motivates the development of an explainable, ETW-driven *Catan* agent capable of both effective gameplay and transparent reasoning. The proposed approach combines interpretable utility-based planning with evolutionary optimisation, while grounding explanations directly in the same decision-making process used by the agent itself. Beyond gameplay performance, the project additionally explores how such reasoning can support tutor-style guidance, post-decision feedback and strategic reflection for human players. Collectively, this work aims to provide insight into explainable decision-making and interactive tutoring in complex stochastic environments.

Chapter 3

Building a *Catan* Tutor

3.1 Prototype Game Environment

To support the development of an explainable *Catan* tutor, a complete game environment was implemented that simulates the board, rules and player interactions according to the official game specification [7]. The environment exposes all legal actions available at a given state and allows both human players and AI agents to participate in full games.

The system is structured using a Model–View–Controller architecture [27]. The game model represents the state of the board and players, the controller manages game flow and rule enforcement and the view provides an interface for interaction. This separation allows AI agents to operate directly over the model while ensuring all actions remain valid and consistent with the rules.

3.1.1 Interaction Architecture

The overall system structure is shown in Figure 3.1. The `GameController` orchestrates gameplay by coordinating between the game state, player inputs and AI decisions. Human interaction is handled through the view, while AI agents replace user input with automated decision-making.

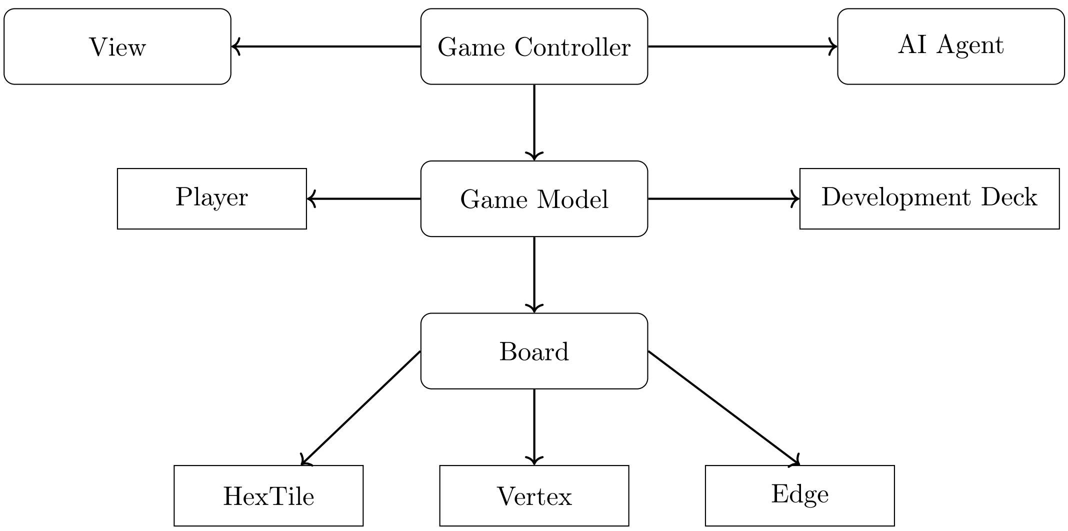


Figure 3.1: High-level architecture of the *Catan* game environment.

3.1.2 Board Representation

The *Catan* board is represented as a 19-tile hexagonal grid, where tiles produce resources, settlements and cities occupy vertices and roads connect edges. A key challenge in modelling the board is that vertices and edges are shared between adjacent tiles, requiring a representation that supports both spatial reasoning and efficient adjacency queries.

To address this, the board is implemented using an *axial coordinate system* [28], where each tile is identified by coordinates (q, r). This provides a compact and consistent way to navigate the hex grid and compute neighbouring tiles. The resulting layout is illustrated in Figure 3.2.

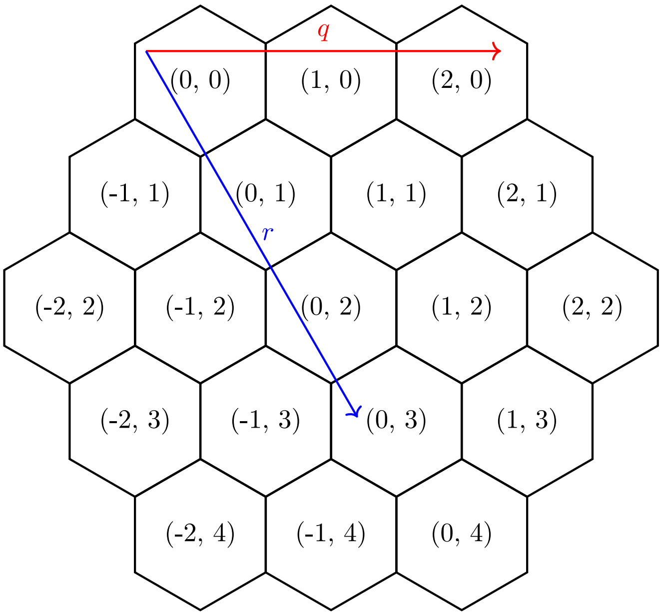


Figure 3.2: 19-tile *Catan* board layout with axial coordinates (q, r) for each tile.

Each tile maintains references to its associated vertices and edges, allowing the board to be treated as an interconnected graph structure. This representation is particularly important for reasoning about connectivity, such as determining valid settlement locations or computing the longest road. While the underlying geometry is implementation-specific, the resulting abstraction exposes a clear interface for querying legal actions and spatial relationships.

Initialisation and Resource Distribution

At the start of each game, tiles are assigned resource types and number tokens according to the official rules, with positions randomised to generate varied board configurations. Ports are similarly distributed around the board edges with spacing constraints.

This randomised initialisation introduces stochasticity into the environment while preserving structural constraints, ensuring that each game presents a distinct strategic landscape.

3.1.3 Player and Game State

Each player is represented by a `Player` object that encapsulates resources, built structures, development cards and victory points. Resource transfers are mediated through a shared bank to ensure consistency and prevent invalid states.

Victory points are tracked as both *visible* (from settlements, cities and achievements such as Longest Road) and *hidden* (from development cards). This distinction is important for modelling imperfect information and opponent reasoning, as players must make decisions based on incomplete knowledge of others' true scores.

Development cards are implemented as a finite shuffled deck, introducing both stochasticity and hidden information. Newly purchased cards are temporarily unplayable to enforce turn-based constraints.

3.1.4 Rule Enforcement and Action Space

The `Game` class acts as the central authority for rule enforcement. All actions, including building, trading, dice rolls and robber movement, are validated and executed through this layer. This ensures that the game state remains consistent and that illegal actions are systematically rejected.

A key feature of the environment is the explicit modelling of the **legal action space**. At any state, the system computes all valid actions available to a player, such as buildable settlement locations, reachable road placements, playable development cards and feasible trades. This allows AI agents to operate over a constrained and well-defined decision space, simplifying decision-making and ensuring that only legal actions are considered. It also avoids duplicating rule logic inside the agents themselves, so decision-making can focus entirely on selecting among feasible options.

This representation is particularly important for a game such as *Catan*, where many actions are only conditionally legal. For example, settlement placement depends not only on the distance rule, but also on connectivity to the player's road network, while road placement depends on existing ownership and blocked paths. By exposing these possibilities directly, the environment effectively converts the raw game state into a structured decision problem.

Several global mechanics are also handled centrally. Resource distribution follows the official rules, including the case where the bank does not have enough resources to satisfy all production. Similarly, *Longest Road* is treated as a dynamic graph property of the current board state. After each road placement, the relevant road network is re-evaluated using graph traversal, allowing ownership of the achievement to be gained or lost as new roads and blocking settlements are added.

Game termination is triggered when a player reaches the victory point threshold, including hidden points from development cards. Centralising these mechanics ensures that all downstream components, including AI agents and later tutoring logic, operate over a valid and fully synchronised game state.

3.1.5 Game Flow and Control

The `GameController` manages the progression of the game, including turn sequencing, dice rolls, building phases, trading and special actions such as robber placement. It acts as the bridge between the game model and external agents.

The controller delegates decision-making to either human input (via the view) or AI policy modules, depending on the player type. This design allows different decision-making strategies to be integrated and evaluated without modifying the underlying game logic.

3.1.6 Interface and Interaction

An initial command-line interface (CLI) was developed to validate game functionality and enable rapid testing of gameplay mechanics. The CLI supports full interaction, including building, trading and special actions, while providing a clear textual representation of the board and player state.

Although functional, the CLI is primarily used as a development tool. A graphical interface was later introduced to improve usability and better support interactive tutoring.

3.1.7 Baseline AI Implementation

A modular baseline AI was implemented to enable automated gameplay and validate end-to-end system behaviour. The AI operates in several modes corresponding to different stages of a turn, including initial placement, standard build decisions, trading, development card usage and robber handling.

This separation was useful during prototyping because different parts of *Catan* involve qualitatively different decision problems. For example, settlement placement requires spatial evaluation of the board, trading depends on current resource deficits and surpluses, while robber placement requires lightweight opponent targeting. Structuring the AI around these contexts made it possible to test the full game loop before introducing more sophisticated strategic planning.

Decision-making is based on simple rule-based heuristics combined with stochastic selection among valid options. In particular, the baseline agent:

- Selects among feasible build locations and actions using lightweight heuristics.
- Performs simple resource-based trading and development card decisions.
- Handles special cases such as robber movement and forced discards.

While this behaviour is strategically limited, it provides a flexible testing framework for validating rule enforcement, turn flow and AI integration. Importantly, this baseline agent is not intended to represent the final tutor system, but rather to establish the infrastructure required for more sophisticated rule-based reasoning.

3.1.8 Summary

The prototype environment provides a complete and modular implementation of *Catan*, supporting both human and AI gameplay. Key design features include a clear separation between state representation and rule enforcement, explicit modelling of legal actions and a flexible architecture for integrating AI agents.

These properties are essential for the development of an explainable tutor. By exposing structured game state and decision points, the environment enables reasoning processes to be traced, analysed and ultimately communicated to the player. This forms the foundation for the strategic planning and explanation mechanisms introduced in later sections.

3.2 Graphical Interface Implementation

To improve usability and interactivity, the game was transitioned from a CLI to a graphical user interface (GUI). This allows players to visualise the board state directly, interact with game elements by clicking and receive immediate feedback on their actions. The GUI preserves the underlying game logic and controller structure while introducing an event-driven view layer using PyQt6 [29].

3.2.1 GUI Architecture

The main GUI architecture consists of a `QtView` class, which acts as a bridge between the `GameController` and the visual components, as illustrated in Figure 3.3. The board is displayed using a custom `SquareCanvas`, while game information, player resources and available actions are presented in a side panel. Communication between the view and controller is managed using Qt's signal-slot mechanism, allowing user input to be handled asynchronously while preserving the sequential structure of turn-based gameplay.

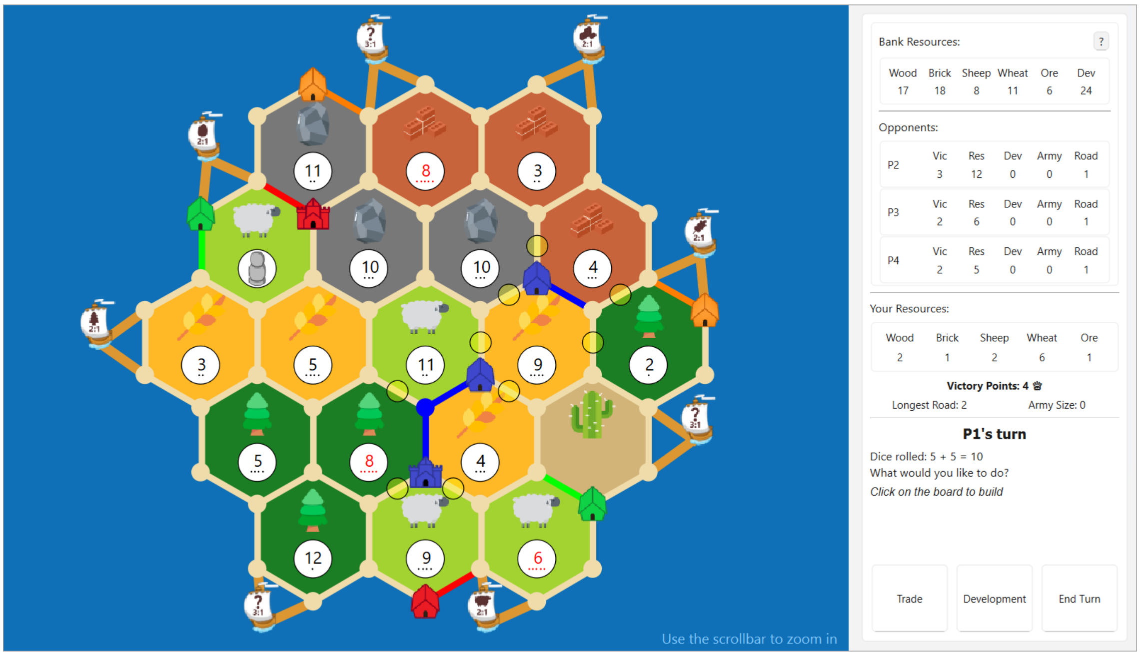


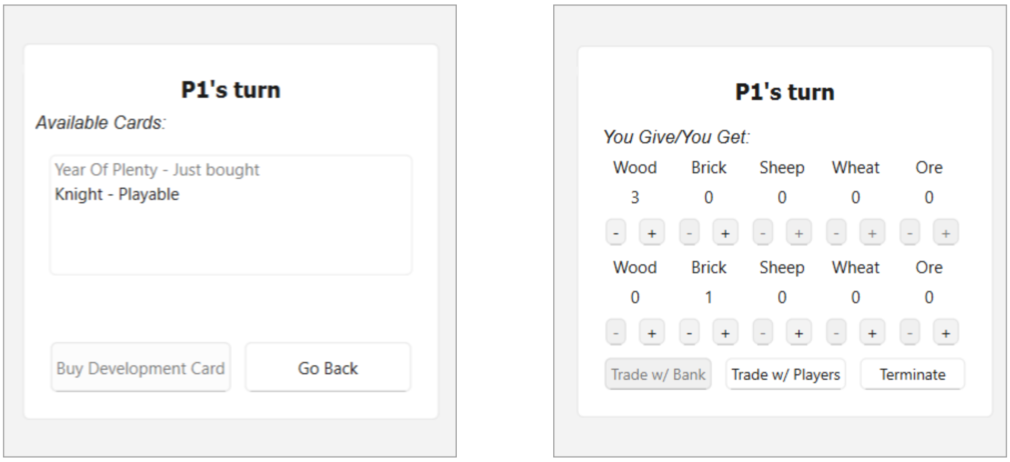
Figure 3.3: Screenshot of the main turn interface. The board is shown, with the right panel presenting player information, game state and available actions.

This architecture maintains a clear separation between presentation and game logic. The controller is responsible for validating and executing actions, while the GUI handles rendering and input collection. This allows the same system to support both human and AI players without duplicating logic.

3.2.2 Interactive Board and Menus

The board is used as the primary interaction surface. Players interact directly with visible elements such as vertices, edges and tiles, rather than referring to them indirectly. This supports spatial decision-making, such as selecting build locations or placing the robber.

Interactive elements include clickable vertices and edges for settlement, city and road placement, along with menus for structured actions such as trading, development card usage and resource selection. Figure 3.4 shows examples of these interaction screens.



(a) Development card management

(b) Trading interface

Figure 3.4: Example screenshots of player interaction menus.

A blocking selection mechanism is used to pause execution until a player completes a required action. This preserves a clear turn-based flow while allowing the interface to remain responsive.

The interface supports AI-controlled players by updating the board and side panel as actions are performed. This allows human and AI players to operate within the same interaction framework.

3.2.3 Mobile Deployment

The separation between the game logic and interface layers also allowed the application to be deployed on Android devices with relatively few code changes. Most of the existing PyQt codebase was reused directly, with modifications primarily limited to adapting large desktop-oriented menus into smaller touch-friendly interfaces. Figure 3.5 shows the tutor system during gameplay on an Android device.

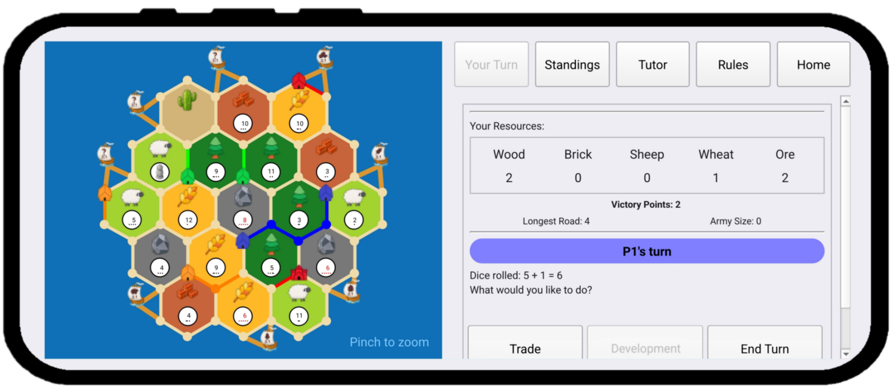


Figure 3.5: Android version of the tutor system during gameplay.

Although a mobile version was successfully implemented, the desktop interface was used throughout development and evaluation, as it provides greater screen space for displaying the game state, tutor recommendations and explanatory information simultaneously.

Overall, the GUI improves usability while maintaining a clean separation between interface and game logic. Its design supports extension for additional features and deployment across multiple platforms without modifying the core game implementation.

3.3 Strategic Planning Agent Development

This section describes the design and implementation of the ETW-driven planning framework used by the tutor system. The architecture combines weighted strategic evaluation, simulation-supported evaluation and utility-based action selection within an interpretable decision-making pipeline. The agent operates by evaluating a set of weighted decision rules that encode domain knowledge about effective gameplay. These rules are applied to the current game state to select actions that maximise progress towards winning.

3.3.1 Decision Framework Overview

The strategic evaluation system defined in Section 2.4 is implemented as a deterministic ETW-driven decision architecture combining counterfactual simulation, utility-based action selection and heuristic planning. The overall objective is to minimise ETW by reducing ETB for strategically valuable actions, while incorporating opponent interference through weighted utility terms.

At each decision point, the current game state is projected into a lightweight counterfactual simulation layer, allowing candidate actions to be evaluated without modifying the true game state. Candidate actions spanning construction, trading and development-card usage are evaluated within this counterfactual planning layer using ETW and utility-based scoring. In practice, candidate generation is deliberately bounded to ensure feasible evaluation while retaining strategically relevant options.

Each candidate is treated as a short plan, but only the first action of the highest-scoring plan is executed, yielding a receding-horizon decision process. Plan evaluation proceeds by simulating the candidate in a counterfactual state, re-estimating ETW using a forward-planning procedure and converting the resulting change into a normalised self-utility term. Where enabled, additional utility components account for opponent delay and progress towards special objectives such as Longest Road, with overall utility discounted by the plan's ETB.

If the selected action is not immediately affordable, the agent may insert a single trade step before execution. Deterministic maritime or bank trades are prioritised, while player trades are proposed or accepted only when they improve the agent's outcome relative to its BATNA (Section 2.4.5).

3.3.2 Rule Design and Heuristics

The rule set is designed to encode high-level strategic preferences rather than prescribing fixed action sequences. Instead of relying on hard-coded priorities, the agent evaluates actions through a collection of heuristics that capture domain knowledge about effective play and long-term planning.

These heuristics can be grouped into several conceptual categories. *Progression heuristics* favour actions that directly reduce ETW, such as constructing settlements or cities, or unlocking future victory point opportunities. *Economic heuristics* prioritise actions that improve resource acquisition, build throughput, or trading flexibility, thereby supporting future progression. Finally, *Interaction heuristics* capture competitive considerations by rewarding actions that delay opponents or secure contested objectives, such as Longest Road.

This decomposition allows multiple strategic considerations to influence decision-making simultaneously. By expressing each heuristic as a soft preference rather than a strict rule, the agent avoids brittle rule ordering and can smoothly trade off short-term gains against longer-term objectives.

3.3.3 Utility Evaluation and Action Selection

Action selection is formulated as a utility maximisation problem, bringing different heuristic signals onto the same numerical scale for comparison. This allows the decision architecture to compare qualitatively different strategic actions within a unified evaluation framework.

For each candidate plan, heuristic contributions are aggregated into a scalar utility value, with positive contributions reflecting progress towards the agent's objectives and negative contributions reflecting opportunity cost or delayed execution. Discounting based on ETB ensures that actions with delayed payoff are penalised relative to immediately effective alternatives, encouraging timely progression.

Given an identical game state, the resulting utility formulation yields deterministic action selection. This property improves reproducibility and supports later explanation, as individual decisions can be traced back to specific utility components rather than stochastic variation.

3.3.4 Difficulty Scaling

To support players with different experience levels, the system includes configurable AI difficulty settings based on epsilon-greedy action selection. At higher difficulty levels, the agent deterministically selects the highest-utility action. Lower difficulty levels increase the probability of selecting a random legal action instead, producing weaker and less consistent gameplay behaviour while retaining the same underlying strategic evaluation framework. This approach is conceptually similar to ϵ-greedy exploration strategies commonly used in reinforcement learning to balance optimal decision-making with behavioural variability [30].

3.3.5 Optimisation for Runtime Efficiency

A naive implementation of the simulation-supported ETW evaluation pipeline would be computationally expensive, as each decision requires evaluating multiple candidate plans via hypothetical rollouts. To ensure responsive gameplay, several optimisations were introduced to reduce both the branching factor and evaluation cost.

First, the agent avoids deep-copying the full game state by using a lightweight counterfactual representation. The real `Game` object is treated as read-only, while hypothetical actions are applied to a `BoardOverlay` storing only ownership deltas alongside `SimPlayerState` copies.

Second, repeated computations are memoised. ETB and ETW estimates are cached using keys derived from relevant parts of the simulated player state and evaluation settings. Candidate generation is also cached to avoid regenerating equivalent action sets within a turn.

Third, candidate generation and rollout evaluation are explicitly bounded. Expansion search uses upper bounds on settlement candidates together with a bounded beam search, while utility evaluation limits the number of candidates scored per decision and discards plans whose ETB exceeds a threshold.

Finally, the agent employs a two-stage **"cheap-first, expensive-later"** evaluation strategy. Lightweight heuristics are first used to rank candidate actions, after which full ETW evaluation is applied only to a small subset of promising candidates.

3.3.6 Interpretability and Explainability

The decision architecture is inherently interpretable, as decisions are produced through an explicit utility decomposition rather than a learned black-box model. Each selected action can be explained in terms of its contributing heuristic components, such as reductions in ETW, economic improvement, or opponent delay, all of which are directly observable and human-readable.

Because action selection is deterministic and grounded in a small number of interpretable utility terms, explanations can be generated by inspecting the relative contributions of these terms for the chosen action. This enables decisions to be justified in terms of *why* an action was preferred over alternatives, rather than merely reporting the action itself.

3.3.7 Baseline Performance Comparison

To establish a baseline for evaluating the proposed ETW-driven agent, its performance was compared against three simpler agents with progressively weaker decision-making capabilities. Each configuration was evaluated over 1000 simulated games and performance is reported as the percentage of games won by the ETW-driven agent, as summarised in Table 3.1.

The implemented ETW-driven agent was evaluated against four baseline opponent configurations:

- **Random**: selects all actions uniformly at random, including placement, trading and building decisions and serves as a lower-bound reference.
- **Myopic Heuristic**: an earlier baseline implementation that follows a reactive, stochastic policy guided by shallow heuristics such as weighted build preferences, round-dependent trading ratios and opportunistic building, without forward planning or long-term optimisation.
- **Random (shared initial placement)**: identical to the Random agent, but using the same initial settlement placement heuristic as the ETW-driven agent.
- **Myopic Heuristic (shared initial placement)**: identical to the Myopic Heuristic agent, but with initial settlement locations selected using the ETW placement heuristic.

Opponent Agent	ETW-Driven Agent Win Rate (%)
Random	100%
Myopic Heuristic	96%
Random (shared initial placement)	84%
Myopic Heuristic (shared initial placement)	62%

Table 3.1: Win rate of the ETW-driven agent against baseline opponents over 1000 simulated games. Values are rounded to whole percentages. Due to stochastic variation, they should be interpreted as approximate performance indicators.

Results from Table 3.1 show that the ETW-driven agent consistently outperforms all baseline opponents. Against purely random behaviour, the agent achieves a perfect win rate, while performance remains strong against the Myopic Heuristic agent despite its use of domain-specific rules.

When initial settlement placement is controlled, win rates decrease for both baselines, indicating that opening position accounts for a substantial portion of overall performance. However, the ETW-driven agent retains a clear advantage even under shared placements, demonstrating that its ETW-driven planning and utility-based action selection provide significant benefits beyond favourable initial conditions.

3.3.8 Limitations of the Rule-Based Approach

Despite its strong performance against baseline opponents, the rule-based agent has inherent limitations. Its behaviour is governed by manually designed heuristics and fixed strategy weights, which limit adaptability and make performance sensitive to these design choices.

These limitations motivate the next stage of this work, which focuses on optimising the strategy weights used in utility evaluation. This aims to improve robustness and performance while preserving the interpretability of the rule-based framework.

3.4 Evolutionary Optimisation of Strategy Weights

This section describes how evolutionary optimisation was used to refine the parameters of the tutor's strategic evaluation framework while preserving the interpretability of the underlying decision architecture. Rather than modifying the decision structure itself, the optimisation process adjusts the relative influence of existing strategic features through simulation-based search.

Table 3.2 lists a subset of representative strategy weights optimised during evolutionary search. These parameters span multiple aspects of gameplay, including settlement placement, special objective progression and risk management.

Weight	**Role**	**Range**
INIT_PLACE_YIELD	Settlement placement yield importance	$[0, 5]$
BUILD_SELF_UTILITY	Own action utility weight	$[0, 5]$
LR_DISTANCE	Progress toward Longest Road	$[0, 3]$
LA_PHASE	Largest Army timing factor	$[0, 3]$
TIME_DISCOUNT_RATE	Preference for short-term gains	$[0, 2]$
ROBBER_OWN_HEX_PENALTY	Avoid self-blocking	$[0, 1]$

Table 3.2: Representative subset of the 46 strategy weights optimised during evolutionary search.

3.4.1 Motivation for Evolutionary Weight Optimisation

Manually tuning a large set of interacting weights is difficult in stochastic multi-agent environments. The effect of a single parameter depends on dice outcomes, opponent behaviour and the current game state, making systematic adjustment through trial and error inefficient. Performance must be evaluated over many simulated games and small parameter changes can lead to unpredictable behavioural shifts due to interactions between features.

Gradient-based optimisation is not suitable in this setting, as performance is measured through discrete gameplay outcomes rather than a differentiable objective function. The resulting search space is noisy and non-smooth, preventing reliable gradient estimation. Evolutionary optimisation therefore provides a practical alternative, allowing candidate weight configurations to be evaluated directly through simulation and iteratively refined based on observed performance. Rather than learning new representations, the objective is to calibrate the balance between existing strategic components.

3.4.2 Initial Evolutionary Approach

The initial optimisation attempt used a simple mutation-based evolutionary loop to tune the strategy weights. Candidate configurations were evaluated through simulated matches, with higher-performing individuals retained and new candidates generated through stochastic mutation within fixed parameter bounds. This baseline approach provided a straightforward method for exploring the continuous search space without requiring gradient information.

In practice, optimisation proved highly unstable due to the stochastic nature of gameplay. Performance measurements varied significantly between evaluations, making it difficult to distinguish genuine improvements from random variation. The problem was compounded by the large search space: attempting to optimise a high-dimensional set of 46 weights resulted in noisy updates and inconsistent progress across generations. As a result, the evolutionary loop struggled to converge toward reliable behavioural improvements.

These issues motivated a revision of the optimisation framework. In particular, the game model was redesigned to support fixed random seeds, ensuring that candidate strategies could be compared under consistent game conditions. This change reduced evaluation noise and enabled more stable optimisation. In addition, the optimisation procedure needed to be replaced with a more structured evolution-strategy approach better suited to high-dimensional continuous parameter spaces.

3.4.3 Optimisation Framework

Following the limitations observed in the initial evolutionary loop, the optimisation process was restructured as a continuous evolution-strategy framework designed for high-dimensional parameter spaces. Each candidate strategy is represented as a vector of 46 weights, forming a point in a continuous search space. Parameter ranges were constrained to maintain stable gameplay behaviour.

Evolutionary search was performed using CMA-ES, introduced in Section 2.7. CMA-ES samples candidate solutions from a multivariate Gaussian distribution whose covariance matrix adapts over time, enabling coordinated adjustments between related weights. This is particularly important given the interdependence between strategic parameters, where meaningful behavioural changes often arise from combinations of weights rather than isolated updates.

Population updates are guided by elitist selection, with higher-performing candidates shaping the sampling distribution for subsequent generations. Instead of recombination, new candidates are drawn directly from the evolving distribution, allowing gradual refinement while maintaining exploration. Combined with fixed evaluation seeds, this framework provided a more stable and scalable optimisation process than the initial mutation-based approach.

3.4.4 Evaluation Protocol

Candidate strategies were evaluated through repeated simulations against a fixed set of opponent policies. To control for game stochasticity, each generation used a predefined set of random seeds and all candidates were evaluated on the same seed set. For each seed, the candidate was tested from multiple starting positions to reduce turn-order and initial placement bias, which can strongly influence early resource access and expansion.

Performance was summarised using a composite fitness score. Candidates received a primary reward for winning, with a secondary shaping term based on the candidate's victory points relative to the mean victory points of the opposing players. This provides a less sparse signal than win-rate alone and rewards strategies that make consistent progress even when they do not win.

In addition to training evaluations, a separate validation stream used an independent seed schedule to check that observed improvements were not specific to the training seed set.

3.4.5 Analysis and Outcomes

Evolutionary search produced a stable set of strategy weights that improved overall gameplay performance while preserving the underlying decision architecture. Across evaluation sets of 1000 simulated games against the Myopic Heuristic (shared initial placement) baseline described in Table 3.1, the evolved configuration increased average win rate from approximately 62% to 70%. This indicates that evolutionary optimisation produced measurable behavioural improvements while preserving the underlying rule-based structure.

Figure 3.6 shows the eigenvalue spectrum of the final CMA-ES covariance matrix. The absence of a sharp drop-off suggests that optimisation did not converge toward a single dominant behavioural direction. Instead, performance gains emerged through small adjustments distributed across many parameters, reflecting the multi-objective nature of *Catan* strategy where placement, timing and opponent interaction must be balanced simultaneously.

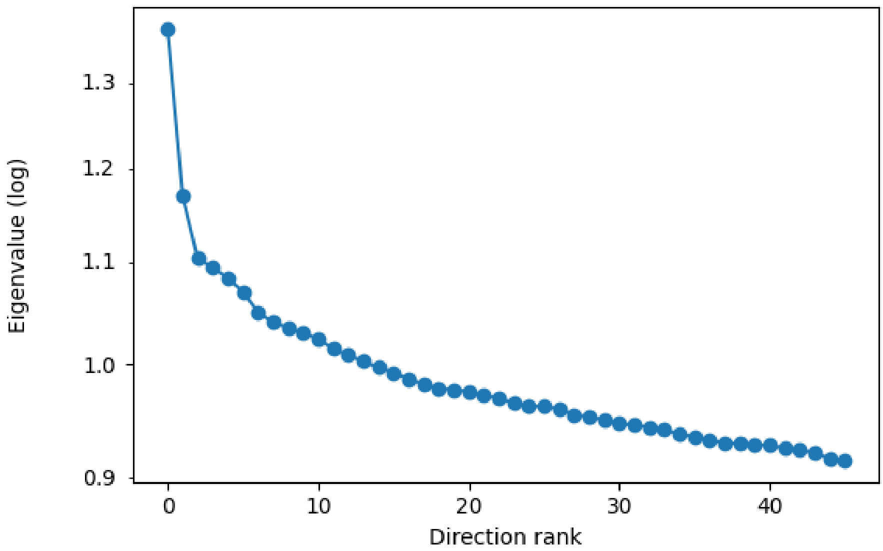


Figure 3.6: Eigenvalue spectrum of the CMA-ES covariance matrix after evolutionary search.

Correlation analysis revealed only weak dependencies between most weights, suggesting that the rule-based design largely separates strategic concerns. A small number of meaningful patterns were nevertheless observed. Time-discounting behaviour showed mild coupling with development-card thresholds, indicating coordinated tuning between immediate actions and delayed advantages. Opponent-interference weights also displayed positive relationships with victory-point gap parameters, implying that the evolved agent became more reactive in competitive board states.

Behaviourally, evolutionary search shifted several strategic priorities in ways consistent with effective *Catan* play. Initial settlement placement moved away from resource diversity and toward stronger blocking behaviour, suggesting that positional control provides greater long-term value than balanced resource access. The evolved agent also increased emphasis on interfering with leading opponents, reflecting a more targeted competitive strategy rather than uniform aggression.

Objective timing also became more differentiated across game phases, with Longest Road treated as a consistent background goal while Largest Army acquisition became more dependent on game phase. In addition, the time discount parameter increased substantially, indicating a stronger preference for immediate gains and faster progression. Across many parameters, values converged toward moderate ranges, implying that evolutionary search primarily refined coordination between existing heuristics rather than driving extreme behavioural changes.

In summary, evolutionary search refined the balance between strategic heuristics and produced a more consistent gameplay policy under stochastic conditions. The resulting weight configuration provides a stable foundation for analysing and explaining decision-making, motivating the explanation framework described next.

3.5 Explanation Generation

This section describes how the project moved beyond selecting strong actions to also justifying them in a way that is understandable to a human player. While the strategy system could already identify effective moves, this alone was not sufficient for an explainable AI system. A player also needs to understand *why* a move is recommended, what broader plan it supports and how it relates to the current board state. To address this, the project introduced an explanation system that converts internal decision information into player-facing justifications that are specific, concise and based on the actual reasoning used by the AI.

Here, a *milestone* refers to any action that increases VP, such as building a settlement, a city or achieving Longest Road. *Reaching the next milestone* means working toward one of these outcomes.

3.5.1 Motivation and Design Goals

A key limitation of exposing internal evaluation directly is that it does not match how players naturally think about the game [12]. In *Catan*, decisions are usually understood in terms of plans, such as expanding toward a settlement, upgrading to a city or preparing for a trade. The explanation system therefore needed to move beyond reporting *how much* a move helps and instead explain *how* it supports meaningful game progress.

To do this, explanations were designed to expose the same short- and medium-term plans used internally during decision evaluation, connecting each move to concrete milestones such as settlement expansion or city progression. This made recommendations more interpretable as part of a coherent sequence rather than as isolated decisions.

Explanation	Assessment
*"Build a **ROAD** because it is the action that reduces your **estimated time to win** the most. This is because it reduces your **estimated turns to win** from **96.2** to **40.4**."*	✗ **Bad** Too literal; exposes internal scoring rather than game reasoning.
*"Build a **ROAD** because it opens a path toward a future **SETTLEMENT**. This is a **medium-term plan**, but it creates a strong **expansion opportunity** and improves your **long-term position**."*	✓ **Good** Player-focused; explains plan and value.

Table 3.3: Comparison of explanation quality for the action **Build a Road**.

This highlights a key distinction between decision quality and explanation quality. While both examples describe the same action, only the latter translates the AI's reasoning into a form that aligns with human understanding of the game.

3.5.2 Explanation Representation

To support structured and consistent explanations, the system represents each decision using an intermediate explanation model. Each candidate action is paired with a `CandidateExplanation`, which stores the full planned sequence of actions to reach the next milestone, expected resource requirements and a set of scored reasons describing why the action is beneficial.

Reasons are stored as labelled categories, such as improving production or enabling expansion. This allows the system to preserve the main decision factors while improving reusability across different explanation types. Multiple candidates are then compared and the selected action is wrapped in an `ActionExplanation` object, which also stores alternative options and a confidence measure.

Natural-language explanations are generated as a final step using predefined explanation structures that convert the stored decision information into player-facing text. These structures vary depending on the action type and available plan information, allowing the system to produce explanations that describe both the immediate move and its role within a broader sequence of actions.

Importantly, explanations are not generated independently of the decision-making process or produced through post-hoc text generation. Instead, they are derived directly from the same planning structures, utility evaluations and strategic reasons used during action selection. This ensures that explanations remain faithful to the agent's underlying reasoning process rather than providing generic justifications detached from the actual decision.

This representation provides a clear separation between decision-making and explanation generation, whilst remaining flexible enough to support different levels of detail and interaction for future tutor-style guidance.

3.5.3 Explanation Presentation and User Interaction

Presenting an explanation effectively required more than text generation. The recommendation also needed to be presented in a way that helped the player connect the explanation to the current board state and understand how the move fits into a wider plan.

In the example shown in Figure 3.7, the recommended action is to **End the Turn**. This is a useful example because the move is not strong on its own. Without explanation, it may appear passive or unhelpful. However, the system explains that ending the turn is preferable because the player is currently saving toward a stronger plan.

Figure 3.7: Example of explanation delivery in the interface. The left panel presents the recommended action and its justification, while the board highlights the planned build sequence visually.

The explanation justifies the immediate recommendation by identifying the next concrete action after waiting, “build a ROAD”, and then explains how this supports the broader milestone of building a “SETTLEMENT”. This makes the recommendation more understandable as part of the AI’s wider game plan. The explanation also identifies the missing resources, “1 WOOD and 1 BRICK”, making it clearer what the player should now be hoping to collect or trade for.

The final justification, that the move “opens up future expansion” and “strengthens progress toward Longest Road”, is generated from the scored strategic reasons attached to the candidate action. This allows the player to see the key strategic factors behind the recommendation rather than only the final choice.

The visual presentation reinforces this further. The explanation panel on the left communicates the reasoning in text, while the board highlights the intended plan directly. In this example, the blue road segment shows the recommended next build and the connected highlighted location indicates the future settlement target. This was especially valuable because describing future build locations through text alone was often difficult to follow. However, also showing the plan spatially on the board helps direct the player’s attention to the relevant part of the position, which is valuable for both explanation and future tutor-style guidance.

3.5.4 Challenges, Limitations and Summary

One challenge was that some actions were naturally more straightforward to explain than others. Actions such as building a settlement or upgrading to a city have an obvious strategic purpose, whereas actions such as ending the turn, discarding resources or choosing between similar trades require more forward planning and therefore need stronger justification. Addressing this required explanations to make future intent more explicit, for example, by exposing the next planned step, the broader milestone and any missing resources.

A second challenge was balancing **detail** against **readability**. More detailed explanations can better reflect the AI’s reasoning, but they also risk becoming too long or disruptive during live play. Related to this, concise explanations can hide uncertainty, alternative plans or the fact that multiple actions may have been similarly strong. The current system therefore prioritises clarity and flow over full strategic completeness.

Explanation quality was also partly dependent on template design. Even when the underlying reasoning was sound, poor phrasing could reduce the clarity and perceived quality of otherwise valid explanations. This meant that explanation quality depended not only on what information was surfaced, but also on how it was expressed.

Overall, the explanation system was substantially simplified by the project’s structured, rule-based decision process. Because plans, milestones and strategic reasons were already represented explicitly, they could be surfaced naturally in player-facing form. At the same time, the limitations identified here also motivate the next stage of the project: moving toward a more interactive tutor, where additional detail, alternatives and follow-up clarification can be provided when needed.

3.6 Tutor Feedback System

While the explanation system made AI decisions understandable, it remained inherently passive. The player could observe what the AI would do and why, but there was no direct engagement with their own decisions. Tutor mode extends this into a more interactive form of guidance, shifting the role of the system from explanation to coaching.

Instead of only presenting recommendations, tutor mode introduces a feedback loop around the player's actions, illustrated in Figure 3.8. Before acting, the player is given a brief indication of what to focus on. After acting, the system evaluates the choice and provides feedback relative to a stronger alternative. This forms a continuous cycle of suggestion, action and reflection, which aligns with established models of experiential learning and feedback-driven improvement [19].

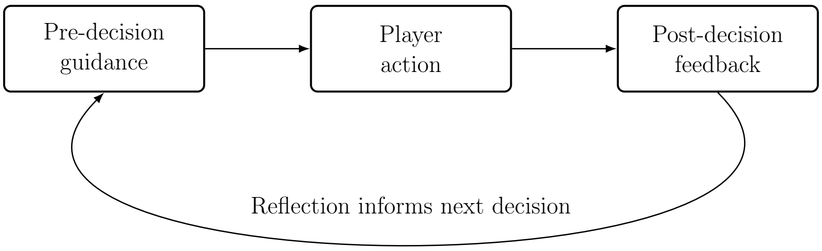


Figure 3.8: Tutor interaction loop illustrating the coaching cycle.

3.6.1 Pre-Decision Guidance

Before the player makes a decision, tutor mode presents the same underlying explanations described in Section 3.5, but in a staged manner rather than immediately revealing the full recommendation. This allows the player to control how much assistance they receive.

At the initial level, the tutor presents general advice about what matters in the current situation. For example, during the opening phase, the panel may display: *"Prioritise strong production numbers, aim for good resource coverage and keep future road expansion in mind."* This encourages the player to reason about the board using general strategic principles rather than directly following instructions.

If the player requests a hint, the tutor reveals a concrete recommendation, such as *"Place a settlement at the field-hill-pasture intersection"*, along with a simple move quality label (e.g. *Excellent*) and a visual indication on the board. This provides a clear reference point for comparison with the player's own idea.

A further explanation can then be requested, expanding the reasoning behind the move. For instance, the tutor may state that it *"improves resource diversity, covers high-frequency numbers and strengthens early production."* This connects the recommendation to core strategic concepts.

This staged approach supports learning by moving from general principles, to concrete examples and finally to explicit reasoning. It allows players to engage at different levels of support, while still encouraging active decision-making.

This separation between decision evaluation and user-facing explanation is conceptually similar to explanation systems used in modern chess platforms, where engine evaluations are translated into interpretable feedback for players [14]. However, unlike post-hoc explanations applied to opaque models, the tutor explanations in this work are grounded directly in the agent's underlying utility and planning process.

3.6.2 Post-Decision Evaluation

When a player makes a move, it is evaluated using the same underlying metrics that the tutor uses to select its own action. In particular, actions are assessed based on their impact on ETW and overall utility within the current game state.

A move quality score is computed for both the player's action and the tutor's preferred action. Comparing these provides a simple indication of how close the player's decision was to the optimal choice, for example, whether it was near-optimal or noticeably weaker.

The same evaluation process is also used to extract the key reasons behind each move. These are then converted into structured feedback, allowing the system to critique the player's decision in terms of its strategic consequences rather than just its score.

Figure 3.9 shows an example of this process. After building a road, the player is shown a comparison such as *"Your move: Build a ROAD. Better move: End the turn."* The feedback then highlights both a positive aspect of the decision, for example *"It opens up future expansion,"* and a key weakness, such as missing the opportunity to save resources to upgrade to a city. This is followed by a short takeaway, e.g. *"Favour the move that advances your best near-term plan,"* guiding future decisions.

Figure 3.9: Post-decision feedback comparing the player's move with the tutor's alternative. The road is shown in faded blue, while the tutor's target city is indicated by a light blue outline.

This approach ensures that feedback goes beyond simply indicating that a move was suboptimal. Instead, it explains how the player's reasoning differs from a stronger alternative, making the underlying decision process more transparent.

3.6.3 Feedback History and Reflection

In addition to real-time feedback, tutor mode stores previous decisions and their evaluations, allowing players to revisit earlier moments in the game and reflect on their choices over time.

This feature is particularly useful because tutor feedback is presented during active gameplay, where players may not always have time to fully read or process explanations before continuing. The history view allows feedback to be revisited afterwards without interrupting the flow of play.

Each entry records both the player's chosen action and the tutor's preferred alternative, together with the associated explanation. The system also restores the corresponding board state, allowing players to step back through earlier positions and reconsider how different decisions may have affected later outcomes.

This supports reflection beyond isolated mistakes by helping players observe how strategic decisions accumulate across the course of a game. By comparing alternative choices in their original context, players can better understand how short-term actions contribute to longer-term positional development and overall game progression.

3.6.4 Limitations of Tutor Feedback

Although tutor mode extends the system from explanation into coaching, its feedback remains dependent on several design assumptions. The following limitations are particularly important when interpreting the tutor's advice.

Action Selection vs Player Move Evaluation While the agent can identify a preferred action, tutor mode must also judge an alternative chosen by the player. This introduces a separate evaluation layer, where the player's action is compared against the tutor's choice and feedback is generated from the differences. As a result, feedback quality depends not only on the strength of the tutor's move, but on how well the system can assess reasonable alternatives. A player's move may be valid even if it differs from the tutor's and overly rigid comparison can lead to misleading criticism.

Dependence of Feedback on Underlying Evaluation Tutor mode operates as a pipeline from game state to feedback, as illustrated in Figure 3.10.

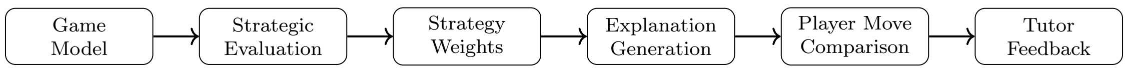


Figure 3.10: Tutor pipeline from game state to final feedback. Arrows show flow.

Each stage depends on the previous one. If the tutor's preferred move is only marginally better, the resulting critique will also be weak. More generally, explanations and feedback can only be as strong as the reasoning they are based on, so limitations in the heuristic evaluation directly affect the quality of tutor output.

The Challenge of Defining Mistakes in *Catan* A key issue when building tutor feedback is deciding what actually counts as a "mistake". In games like chess, this is usually clear, as a bad move can immediately lose material or lead to a forced sequence. In more stochastic games such as *Monopoly*, outcomes are often driven by chance rather than decision quality. For instance, landing on a monopoly with hotels can result in a severe loss that can effectively end a player's game.

Catan falls between these extremes, but its progression is more passive and incremental. Most decisions have delayed effects, with turns often resulting in no immediate gain or only small improvements in production or positioning. The game therefore advances through accumulation rather than decisive moments. As a result, mistakes are difficult to define in isolation: a decision rarely loses the game outright, but instead slows progress relative to alternatives, meaning evaluation must focus on long-term impact and marginal differences rather than objectively incorrect moves.

However, some decisions can still have immediate and significant consequences, particularly trades in the late game. Accepting or offering a trade that directly enables another player to complete their plan, such as building for the win, can effectively decide the outcome of the game, introducing rare but decisive errors. This is also one of the hardest areas to evaluate: a trade may appear weak according to the AI, but still be reasonable given the player's expectations or incomplete information. As noted by Thomas [8], *Catan* AI cannot infer opponent plans through communication, meaning trades that depend on hidden intent may be misjudged.

3.6.5 Post-Game Analysis and Review

While tutor mode provides real-time guidance during gameplay, the post-game analysis supports reflection after the game has finished. It allows players to review their decisions in context and understand how individual choices contributed to the final outcome.

Figures 3.11 and 3.12 illustrate the post-game review system, which combines final results, performance evaluation and tutor feedback to support reflection after gameplay. The system reports an overall performance score that integrates decision quality with the final game outcome, providing a single measure of how effectively the player converted decisions into progress. In addition, a short qualitative summary highlights key strengths and weaknesses, allowing players to quickly interpret their performance without reviewing every individual move.

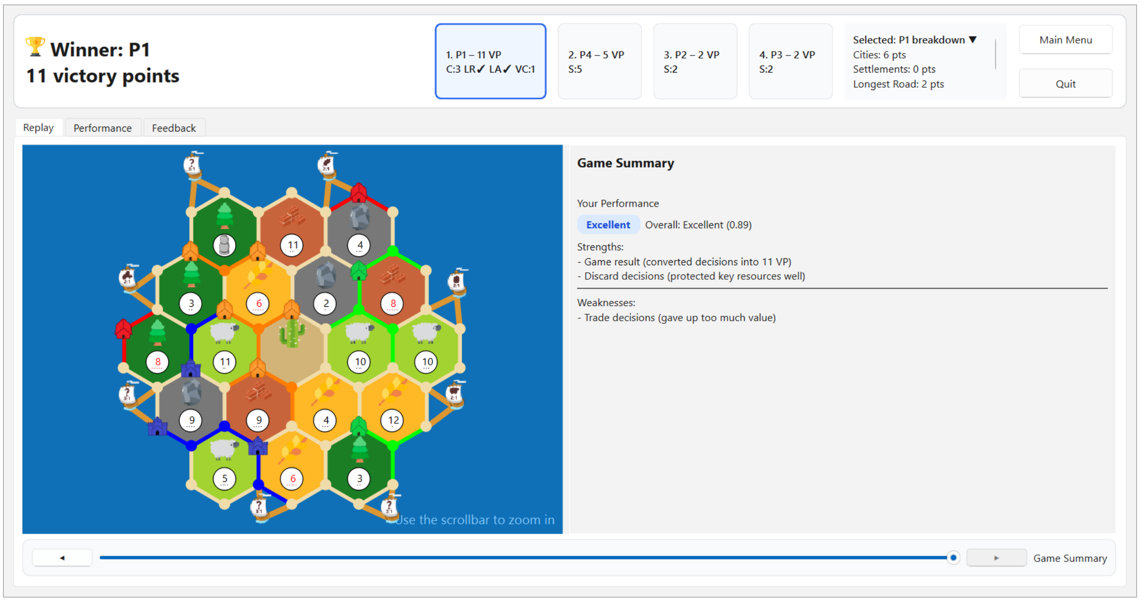
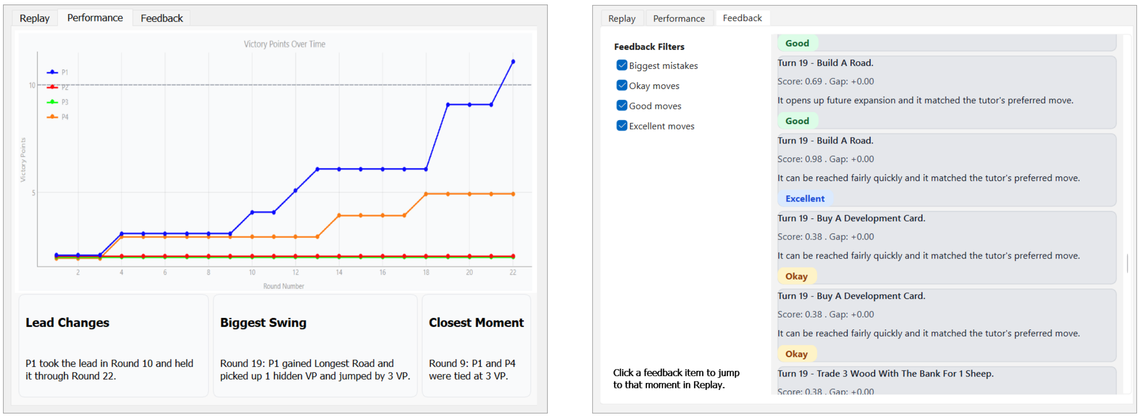


Figure 3.11: End-of-game review showing results, performance evaluation and tutor feedback.

The *Replay* tab (Figure 3.11) supports temporal understanding by allowing players to step through the game timeline using a replay slider and revisit past decisions in context. It builds on the same feedback history available during gameplay, but presents it in a dedicated review setting. By restoring both the board state and associated tutor feedback at each point, it highlights how individual choices accumulate over time, making it easier to identify key turning points rather than viewing moves in isolation.

The *Performance* tab (Figure 3.12a) provides a high-level view of game progression. It visualises victory points over time and summarises key moments, helping players understand how the game evolved and how their decisions translated into long-term outcomes. These insights are intentionally lightweight, offering quick interpretation rather than detailed analysis.

The *Feedback* tab (Figure 3.12b) presents the complete history of tutor evaluations. Moves are organised by quality and can be filtered to focus on strengths or mistakes, with each entry linked back to the corresponding game state. This allows players to review decisions in context and better understand the reasoning behind stronger alternatives.



(a) Performance Evaluation

(b) Feedback Review

Figure 3.12: Post-game analysis showing performance evaluation and tutor feedback.

3.6.6 Summary

The tutor system successfully extends the role of the AI from explanation to interactive guidance. By introducing a feedback loop around player decisions, it encourages active engagement rather than passive observation, aligning with established principles of feedback-driven learning. The combination of pre-decision guidance, post-decision evaluation and the ability to revisit past decisions provides a coherent framework for both immediate improvement and longer-term reflection.

However, the effectiveness of the tutor is closely tied to the quality of the underlying evaluation. In *Catan*, decisions can have delayed and often uncertain outcomes, meaning the distinction between a strong move and a mistake is not always clear-cut. As a result, the tutor is best viewed not as a source of definitive answers, but as a tool for highlighting relative strengths, weaknesses and alternative lines of play.

Overall, the system demonstrates that it is possible to provide meaningful, interpretable feedback in a complex stochastic environment. While there are inherent limitations in evaluation and judgement, the tutor succeeds in making decision-making more transparent and in supporting player learning through both real-time guidance and post-game analysis.

Chapter 4

Evaluation

4.1 Evaluation Overview

This chapter presents a comprehensive evaluation of the implemented *Catan* tutor across four key dimensions: gameplay performance, optimisation effectiveness, explanation quality and tutor usefulness. The aim is to assess not only whether the system plays competently, but also how users interact with its recommendations and explanations, whether they find them useful and what insights emerge regarding trust, understanding and tutor usability.

The evaluation combines quantitative and qualitative methods. Gameplay performance is analysed through large-scale simulation tournaments and ablation studies, measuring the contribution of different components to overall strength. The impact of evolutionary optimisation is examined by analysing how weight tuning affects overall system performance and decision quality and how these improvements contribute to the reliability of tutor recommendations. Explanation quality is assessed using structured analysis of decision-level outputs, while tutor effectiveness is evaluated through a small user study capturing both performance and user experience.

Together, these experiments provide a comprehensive assessment of the system, highlighting its strengths, limitations and suitability as an explainable AI tutor in a complex stochastic environment.

4.2 Gameplay Performance

4.2.1 Overall Tournament Results

To evaluate overall gameplay performance, a series of large-scale tournaments was conducted within the implemented simulation environment. Four agent types were compared: a purely Random agent, a Myopic heuristic agent, the ETW agent with manually defined baseline weights and the ETW agent with evolved weights obtained through CMA-ES optimisation. As discussed in Section 3.3.7, earlier experiments evaluated agents in isolation against multiple identical opponents (e.g. one ETW-driven against three Random agents). While useful for assessing baseline competence, this setup does not allow direct comparison between different agent types under identical conditions.

Each tournament consisted of 1000 simulated four-player games. In every game, one instance of each agent participated, ensuring direct competition under identical conditions. To reduce bias from turn order and its effect on initial settlement placement, player positions were rotated across games and a fixed set of random seeds was used to ensure consistency across evaluations.

Performance was measured using multiple metrics. Win rate captures the primary objective of the game, while average victory points (VP) provide a more continuous measure of performance across all games. The VP gap to second place reflects how decisively an agent wins and average game length (in turns) provides insight into the pace of gameplay induced by different strategies.

The chosen baselines reflect increasing levels of strategic sophistication. The Random agent provides a lower-bound reference, while the Myopic agent captures simple heuristic behaviour without long-term planning. The baseline ETW agent isolates the impact of the decision framework without optimisation and the evolved agent represents the final system used for tutoring.

Agent	Win Rate (%)	Avg VP	VP Gap to 2nd (Wins)	Avg Turns (Wins)
Random	0.0	3.4	N/A	N/A
Myopic	3.3	4.0	+0.6	46.3
ETW (baseline)	43.9	7.5	+0.9	34.4
ETW (evolved)	52.8	7.9	+1.2	35.3

Table 4.1: Tournament performance across agents (1000 games).

The results in Table 4.1 suggest a clear progression in performance as agent sophistication increases. The Random agent does not win any games in this setting and consistently achieves low VP totals. The Myopic agent shows only limited improvement, achieving a small number of wins and slightly higher average VP, but its lack of long-term planning significantly constrains its effectiveness.

The ETW agent with baseline weights achieves a substantial improvement in both win rate and average VP, indicating the effectiveness of the ETW-driven decision framework. Compared to the Myopic agent, it demonstrates a much stronger ability to convert strategic planning into consistent game outcomes.

The evolved ETW agent further improves on this performance, achieving the highest observed win rate and the highest average VP across the evaluated games. Compared to the baseline ETW agent, this represents an improvement of almost nine percentage points in win rate, suggesting that CMA-ES optimisation successfully improved the balance between competing strategic heuristics. It also attains the largest average VP margin in its wins, indicating that its victories tend to be more decisive than those of the baseline ETW agent.

In terms of game length, both ETW agents produce substantially shorter winning games than the Myopic agent, suggesting more efficient progression toward victory. The evolved agent exhibits a similar average game length to the baseline ETW agent, indicating that its performance gains arise primarily from improved decision quality rather than faster gameplay alone.

Overall, these results indicate that the proposed system achieves a strong level of gameplay competence relative to the tested baselines.

4.2.2 Position Sensitivity

Player order plays an important role in *Catan* due to the structure of the initial settlement placement phase. Players acting earlier in the turn order have access to a wider range of settlement locations, while later players can react to earlier placements but may face more constrained choices. This creates a potential imbalance, where both first-move advantage and the ability to adapt to prior placements can influence early resource access and, consequently, long-term outcomes.

To examine the effect of player order, results from the 1000-game mixed-agent tournament described previously were analysed by starting position (P1–P4), corresponding to turn order.

Agent	P1 Win (%)	P2 Win (%)	P3 Win (%)	P4 Win (%)
Evolved Agent	45.1	57.7	46.3	46.5

Table 4.2: Win rate by player position (turn order) for the evolved agent.

The results in Table 4.2 show variation across player positions, although not in the expected monotonic pattern. While earlier positions are often assumed to confer an advantage, the highest win rate is observed in P2, with relatively similar performance across P1, P3 and P4.

This suggests that the relationship between turn order and performance may be more complex than a simple first-player advantage, although the observed differences should be interpreted cautiously, given the stochastic nature of gameplay. In *Catan*, the initial settlement phase follows a snake ordering (P1, P2, P3, P4, then P4, P3, P2, P1), meaning that acting first also results in placing the final settlement. As a result, P2 may represent a more balanced position, combining early access to strong first-settlement locations with a relatively early second placement.

Despite this variation, the differences across positions remain moderate and the evolved agent performs competitively in all roles. This suggests that while turn order has some influence on outcomes, overall performance is not dominated by starting position and is instead driven primarily by strategic decision-making throughout the game.

4.2.3 Robustness Across Seeds

Catan is a stochastic environment, with randomness arising from dice rolls, resource distribution and initial board configuration. As a result, agent performance can vary depending on the sequence of events, making it important to evaluate policies across multiple seed sets.

The training seeds correspond to the fixed set used during evolutionary optimisation, while the validation seeds consist of an independent set of unseen seeds. Both evaluations were conducted using the same tournament setup described in the previous section, with 100 games played on the optimisation seed set and a further 100 games on the independent validation seed set. Although fixed seeds reduce variance during optimisation, performance evaluation remains inherently noisy due to stochastic gameplay and multi-agent interactions.

Seed Set	**Win Rate (%)**	**Avg VP**
Training Seeds	45.6	7.28
Validation Seeds	42.8	7.22

Table 4.3: Performance of the evolved agent across optimisation and independent seed sets.

The results in Table 4.3 show that performance on the validation seeds remains close to that observed on the optimisation seeds, with only a small decrease in both win rate and average VP. This suggests that the evolved weight configuration is not overly dependent on the specific seed set used during optimisation.

The small difference between the two settings suggests that performance is reasonably stable under stochastic variation. Although performance varies slightly across seed sets, the overall ranking of agent behaviour remained consistent across evaluations.

4.3 Ablation Study

4.3.1 Component Contribution

The purpose of this ablation study is to isolate the contribution of individual components within the final agent. An ablation study systematically removes or disables specific components while keeping the remainder of the system unchanged, allowing performance differences to be attributed to core elements of the decision-making process, such as planning, opponent awareness and economic interaction.

All variants were evaluated using the same experimental setup, consisting of one test agent against three baseline ETW agents, with results averaged over 500 simulated games per variant.

Variant	Win Rate (%)	Avg VP
Full System	43.2	7.2
No Player Trading	42.2	7.0
No Forward ETW Planning	41.6	7.7
No Opponent Interference	40.8	7.1
No Development Cards	36.0	6.8
No Time Discount	33.8	6.6

Table 4.4: Ablation results showing the effect of removing key decision components.

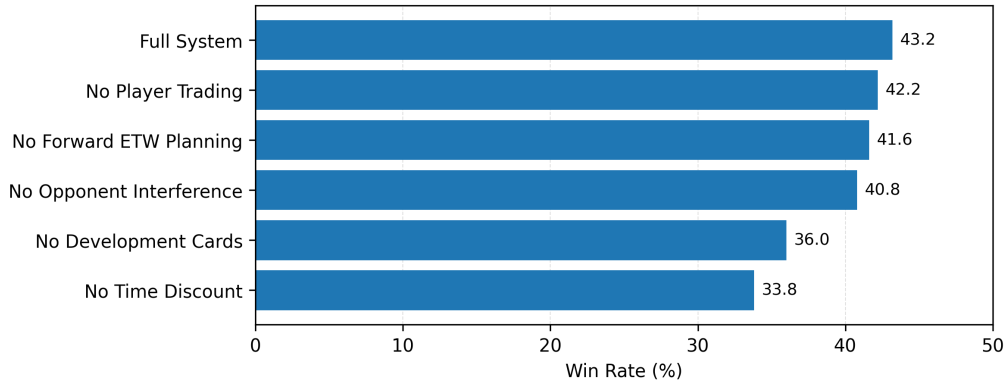


Figure 4.1: Win rates for the ablation study.

The results in Table 4.4 and Figure 4.1 show that component importance varies, with time discounting having the largest impact. Removing it leads to a substantial drop in win rate, providing evidence that prioritising near-term gains plays an important role in converting strategic advantage into wins. Development cards also contribute significantly, as their removal limits access to alternative victory pathways such as Largest Army, as well as hidden victory point cards.

In contrast, removing forward ETW planning results in only a small decrease in win rate, suggesting that much of the agent's strength comes from immediate ETW-based evaluation rather than deeper lookahead. The higher average VP in this variant suggests that while local decisions remain strong, they may be less effectively converted into victories.

Opponent interference and player trading show comparatively smaller effects in this implementation. Disabling either component does not substantially degrade performance, suggesting that the current rule-based heuristics provide only a limited approximation of strategic interaction. In *Catan*, both opponent modelling and trading involve implicit coordination and anticipation of other players' intentions, which are difficult to capture without communication or more advanced modelling.

Overall, the results suggest that the agent's performance is driven primarily by short-term prioritisation and access to multiple strategic pathways, while planning depth and interaction-based components provide more incremental improvements.

4.4 Decision-Level Analysis

4.4.1 Tutor Confidence Across Action Types

In addition to overall gameplay performance, it is important to analyse how the tutor evaluates different categories of decisions. Some actions in *Catan*, such as initial settlement placement, have clear long-term strategic consequences, while others, such as trading decisions, may involve several similarly viable alternatives. Analysing tutor confidence across action types therefore provides insight into how decisively the tutor evaluates different categories of gameplay decisions and where strategic ambiguity most commonly arises. This is particularly relevant for tutoring systems, where the ability to distinguish between clear recommendations and genuinely ambiguous situations may influence how feedback is interpreted by users.

Tutor evaluations were collected across 100 simulated games and grouped by action type. For each decision, the tutor's move-quality score and the preference margin between the highest-scoring and second-highest-scoring legal actions were recorded. Larger preference margins indicate that the tutor strongly preferred a particular action, while smaller margins suggest more ambiguous decision states.

Action Type	Avg Move Quality	Avg Preference Margin
Initial Settlement Placement	0.81	0.06
Settlement / City Build	0.49	0.04
Road Placement	0.53	0.22
Trading Decisions	0.42	0.01
Robber Placement	0.73	0.15
Development Card Usage	0.43	0.43

Table 4.5: Average tutor move-quality scores and preference margins across action types.

The results in Table 4.5 show substantial variation in tutor confidence across different categories of gameplay decisions. Initial settlement placement achieves the highest average move-quality score within the evaluated sample, likely reflecting the more structured nature of opening evaluation, where decisions depend primarily on static board features such as production probability, resource diversity and positional access.

In contrast, trading decisions produce both the lowest move-quality scores and the smallest preference margins, suggesting that these situations are often strategically ambiguous, with several actions providing similar value. Development card usage exhibits the largest observed preference margin despite a relatively modest average move-quality score, indicating that these decisions frequently contain one dominant preferred tactical action. Road placement and robber decisions occupy an intermediate position, reflecting the situational nature of expansion and disruption strategies.

Overall, the analysis suggests that tutor confidence adapts naturally to the structure of the underlying decision. Actions with clearer strategic implications tend to produce more decisive evaluations, while economically and socially complex interactions, such as trading, lead to more moderate assessments.

4.4.2 Strategic Behaviour Over Game Progression

To analyse how the tutor's strategic priorities change throughout a game, tutor-recommended actions were recorded across 100 simulated matches. Since *Catan* games vary in length, each recommendation was normalised by total game duration using the player's progress through the game, measured as the percentage of turns completed relative to the final turn count. Recommendations were then grouped into 11 bins, allowing action frequencies to be compared consistently across different games.

Figure 4.2 illustrates how the tutor's strategic priorities evolve throughout different stages of gameplay. Early-game recommendations are dominated by road construction, reflecting the importance of expansion, network connectivity and access to future settlement locations during the opening phase. Settlement recommendations increase rapidly during the early stages of the game and remain consistently important throughout much of the mid-game, indicating that territorial expansion continues to be strategically valuable beyond the initial opening turns.

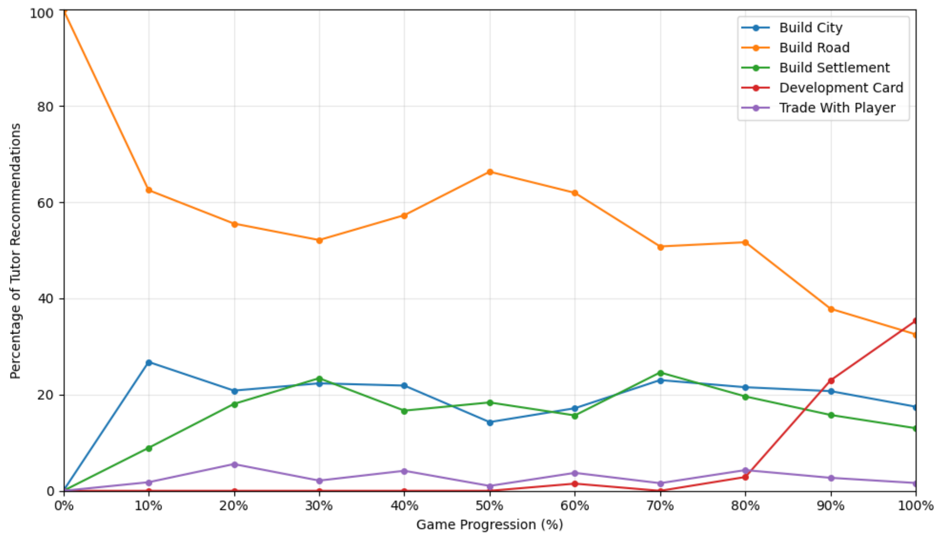


Figure 4.2: Distribution of tutor-recommended action types across normalised game progression.

City-building recommendations emerge earlier and peak more quickly than settlement recommendations, suggesting that the tutor often prioritises upgrading existing settlements before pursuing additional expansion. This reflects a strategic preference for improving existing production locations before pursuing additional territorial expansion.

Development card recommendations increase substantially during the late game and eventually become more frequent than road-building actions near the end of the match. This behaviour is consistent with end-game strategies focused on hidden victory points, Largest Army progression and short-term tactical advantages. In contrast, road-building recommendations decline steadily during later stages as expansion opportunities diminish and the strategic value of additional road placement decreases.

Overall, the results demonstrate that the tutor adapts its recommendations according to game phase rather than relying on a fixed set of priorities throughout the match. The observed progression from expansion-focused play toward consolidation and objective-oriented strategies is consistent with effective human *Catan* play and supports the tutor's ability to provide context-sensitive strategic guidance.

4.5 Explanation Evaluation

4.5.1 Feature Coverage

To evaluate explanation structure, a balanced sample of 100 decision-level explanations was manually annotated (20 from each of five full games) using a binary tagging scheme across four reasoning dimensions: (i) **milestone reasoning** (how the action progresses toward key objectives), (ii) **resource reasoning** (explicit mention of economic constraints), (iii) **future planning** (reference to downstream strategic goals) and (iv) **opponent impact** (consideration of other players' positions or disruption effects).

Feature	Percentage (%)
Mentions Milestone	99
Mentions Resources	74
Mentions Future Plan	72
Mentions Opponent Impact	12

Table 4.6: Frequency of explanation features across sampled tutor decisions.

Table 4.6 shows that explanations are strongly grounded in decision-level reasoning. Almost all explanations connect actions to milestone progression, showing that the tutor consistently frames decisions in terms of how they advance key objectives such as settlements and cities. Resource reasoning is also frequently present, reflecting awareness of economic constraints. Forward-looking planning appears in a large proportion of cases, indicating that actions are typically linked to longer-term goals rather than justified in isolation.

Opponent-related reasoning is less frequent. This is expected, as it is mainly relevant for actions that directly affect other players, such as robber placement. In these cases, explanations explicitly reference disruption or pressure on leading players. Outside these contexts, opponent considerations are integrated into the decision process and are only occasionally stated explicitly when they have a clear strategic impact (see Strong Example 1 below).

These frequencies reflect how different forms of reasoning are expressed across in-game decision contexts, with certain features appearing more strongly in specific types of actions.

4.5.2 Qualitative Examples

This section presents representative examples of the tutor's explanations in order to demonstrate the quality and structure of its reasoning. These examples provide concrete illustrations of how the system justifies its decisions in practice.

Strong Example 1:

<u>Build A Road Here</u>: The plan starts by building a ROAD. After that, the next step is to build a ROAD. That sets up the final step: build a SETTLEMENT. The earlier steps matter because they create access to that settlement opportunity. This is strong because it opens up future expansion, it also slows the current leader, and it can be reached fairly quickly.

Observed features: Milestone, Future plan, Opponent impact

This explanation is strong because it expresses a clear multi-step action chain and explicitly connects early infrastructure to a concrete future objective. Opponent impact is explicitly stated as it is relevant to the decision.

Strong Example 2:

<u>Trade With The Bank</u>: The plan starts by trading 4 WOOD with the bank for 1 WHEAT to save up for a city. That sets up the final step: build a CITY. This is strong because it improves future resource production and it can be reached fairly quickly.

Observed features: Milestone, Resources, Future plan

This explanation is strong because it explicitly links a resource conversion decision to a downstream strategic objective. It also situates the trade as a transitional step toward a higher-value milestone rather than an isolated exchange.

Moderate Example:

<u>Discard These Resources</u>: Discard 1 WOOD and 3 SHEEP. This keeps your stronger follow-up plan available: the next thing we want to build: a settlement. This is strong because it protects the resources needed for your best next plan and it throws away surplus resources first.

Observed features: Future plan

This explanation provides limited strategic grounding because it remains at the level of general resource management without explaining why the discard choice is optimal relative to alternative actions or game-state constraints.

Weak Example:

Place Your Opening Road Here: The idea is to place your road where it keeps your opening flexible. This is strong because it keeps your road network flexible.

Observed features: None

This explanation is weak because it is purely abstract and does not reference any concrete game-state features such as resources, positions, or strategic objectives. As a result, it provides little insight into why the action is preferred over alternatives.

The sampled explanations suggest that the tutor generally produces structured outputs grounded in game-state reasoning. Strong explanations explicitly connect immediate actions to longer-term strategic objectives, often referencing both expansion pathways and resource efficiency. In some cases, they also incorporate opponent-aware reasoning, particularly when describing blocking or disrupting opponent progress.

Weak explanations tend to arise when reasoning is overly compressed, omitting intermediate game-state structure such as resource constraints, milestone progression, or positional consequences. In these cases, actions (e.g. building roads or ending the turn) are often justified in general terms such as flexibility or expansion, without explicitly linking them to concrete resource pathways or follow-up build sequences. Some explanations also become circular or repetitive, restating a recommendation using different wording rather than providing additional strategic justification.

However, overall, the results suggest that the system is capable of producing human-interpretable explanations that consistently expose the key factors underlying decision-making in a structured form.

4.6 Tutor Evaluation

A small-scale user study involving five participants was conducted to evaluate the tutor from a user-centred perspective. The study focused on how players interacted with tutor recommendations and explanations, what aspects they found useful and what limitations emerged during practical use. It was not intended to measure objective skill improvement or long-term educational effectiveness, which would require a substantially larger longitudinal evaluation.

Participants ranged from beginner to intermediate familiarity with *Catan*, with none considering themselves expert players. Each participant completed one standard game without tutor assistance, followed by one game using the tutor-enabled mode. Feedback was collected through questionnaires and detailed participant feedback.

4.6.1 Participant Ratings

Participants completed a short post-study questionnaire using a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). A summary of the results is shown in Table 4.7:

Question	**Mean Rating**
Helped understanding of strategy	4.2
Clarity of explanations	3.2
Usefulness of recommendations	4.6

Table 4.7: Average user ratings of the tutor system.

Overall, participants reported positive experiences with the tutor. Recommendation usefulness received the highest rating, while explanation clarity was rated lower, suggesting that users often found the tutor's recommendations more convincing than the reasoning provided. Four out of five participants also indicated that they would use the tutor again. However, the qualitative feedback proved substantially more informative and is therefore the primary focus of the following analysis.

4.6.2 Strategic Understanding Through Comparison

One of the most notable findings from the qualitative feedback was that participants often valued the tutor's recommendations at least as highly as its explanations. Prior to evaluation, it was expected that explanations would act as the primary educational mechanism, helping users understand the strategic reasoning behind recommendations. However, the detailed playthrough feedback suggested that simply being shown a stronger alternative action was itself valuable for reflection and understanding.

When asked what aspect of the tutor they found most useful, one participant responded:

> "The simple act of pointing out the optimal play for each turn."

This suggests that recommendations provided immediate strategic guidance, while explanations primarily served a supporting role. Participants appeared to develop their understanding by comparing their own decisions against the tutor's preferred alternatives and reflecting on the differences between them.

Figure 4.3 illustrates the process of comparison and reflection that emerged from participant feedback.

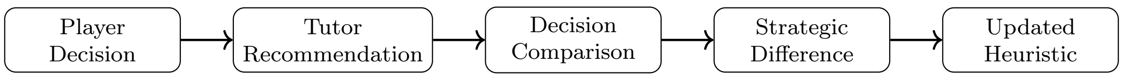


Figure 4.3: Conceptual model of the decision-comparison process observed during tutor interaction.

Several examples of this process were observed.

Resource Management The feedback indicated a change in how discard decisions were approached. Rather than attempting to maintain a balanced distribution of resources, the participants became more willing to discard resources that did not contribute to an immediate strategic objective.

This represents a change in decision-making strategy rather than a one-off correction. Instead of maintaining a balanced resource distribution, resources were evaluated according to their contribution to future plans. This suggests that the tutor influenced an existing heuristic, encouraging a more plan-driven approach to resource management.

Strategic Prioritisation When asked whether the tutor had improved their understanding of *Catan* strategy, one participant highlighted a change in how they viewed city development:

> "My go-to strategy has typically been to build as many settlements and roads as possible, but the tutor is showing me a different idea, to build cities earlier on."

Rather than adopting a specific tactical rule, the participant recognised a recurring strategic preference within the tutor's recommendations. Repeated exposure to city-focused advice appeared to influence their long-term planning, suggesting that the tutor was capable of shaping higher-level strategic thinking. This aligns with Figure 4.2, which shows the tutor favouring city development early in the game relative to settlement expansion.

Opening Settlement Placement The opening settlement placement advice provided another example of understanding developing through comparison. Several participants reported that the tutor identified stronger opening locations than the ones they had originally chosen, helping them recognise more valuable settlement placements.

In these situations, participants selected an opening placement before being shown the tutor's preferred alternative. By comparing the two options, they were able to identify stronger opening positions and reported increased confidence in evaluating settlement placements in future games.

Key Finding Collectively, these examples suggest that much of the educational value provided by the tutor arose from exposing users to alternative strategic decisions and encouraging comparison with their own reasoning. While explanations remained useful for providing context, the strongest evidence of changed strategic reasoning was associated with changes in player heuristics and strategic priorities following repeated exposure to tutor recommendations.

4.6.3 Trust and Perceived Competence

The qualitative feedback suggested that trust in the tutor was primarily driven by observed performance rather than explanation quality. Prior to using the tutor, participants generally acknowledged that *Catan* contains a significant degree of randomness and that following good advice does not guarantee a successful outcome.

During the detailed playthrough, participants reported gaining confidence in the tutor after observing stronger performance than they typically achieved. This included improved opening settlement placement, faster victory point accumulation and a clearer strategic direction throughout the game. Although future outcomes could still be influenced by randomness or the actions of other players, an eventual loss was not considered sufficient reason to distrust the tutor if its recommendations continued to produce strong positions throughout the game.

Key Finding Trust appeared to be established through evidence that the tutor's recommendations were effective, rather than through the persuasiveness of individual explanations. Explanations helped users understand recommendations and reflect on strategic decisions, but confidence in the tutor was primarily associated with observable improvements in gameplay, such as stronger openings, faster progression and more effective strategic planning. This suggests that, within a strategic game such as *Catan*, demonstrated competence may play a greater role in establishing trust than explanation quality alone.

4.6.4 Limitations of the Tutoring Approach

Although the tutor was generally perceived as useful, the qualitative feedback highlighted several limitations that stem from the tutoring approach itself rather than individual explanation templates.

Static Recommendations versus Interactive Tutoring

The current tutor provides recommendations and explanations in a one-directional manner. While this was sufficient for many routine decisions, participants eventually sought clarification beyond the information provided.

> "I hoped the tutor would be able to tell me when best to buy a development card, or when to use a knight... But I couldn't ask specific questions like that... If it allowed me to ask follow-up questions, I would certainly use it to improve my skill at Catan."

This suggests that recommendation-based tutoring may eventually reach a limit. The current system can explain recommendations, but it cannot participate in a dialogue. As users engage with higher-level strategic concepts, they may require clarification, examples, or follow-up questions that cannot be anticipated in advance.

Reactive Planning versus Strategic Continuity

A second limitation emerged from the planner's reactive nature. The tutor continuously re-evaluates the game state and selects the highest-scoring strategy according to current conditions. While desirable from a decision-making perspective, this can conflict with how users perceive strategic planning.

For example, the tutor might encourage progression towards a city, temporarily recommend building a settlement and later return to the city plan. While rational from the planner's perspective, the strategic objective can appear to change without explanation.

This highlights a mismatch between optimisation and tutoring. Players often expect continuity in strategic goals, whereas the planner evaluates each decision independently. Future systems may therefore benefit from referencing previous recommendations when generating explanations, rather than relying solely on the current board state.

Explanation Depth

Participants generally agreed with the tutor's recommendations, but often wanted a deeper explanation of why one strategic option was preferred over another. For example, when the tutor recommended saving resources for a city, participants understood the objective but still wanted to know why this was preferable to alternatives such as building roads or settlements.

This suggests that explanation quality is not determined solely by correctness. Explanations must also provide sufficient depth to communicate the trade-offs between competing actions. While concise explanations improve readability, they may omit details that users consider important for understanding the recommendation.

Key Finding The most significant limitation identified during evaluation was not the quality of individual recommendations, but the scope of the tutoring interaction itself. The system successfully provided strategic guidance and influenced player behaviour, yet users increasingly sought deeper reasoning, continuity across decisions and opportunities to ask their own questions. These findings suggest that future tutoring systems may benefit from moving beyond static recommendation-and-explanation paradigms towards more interactive forms of strategic coaching.

4.6.5 Discussion

Several insights emerged from the evaluation that extend beyond the quality of individual explanations and instead concern the broader design of explainable tutoring systems.

Milestone-Based Planning Successfully Communicates Strategic Goals The clearest changes in player reasoning were associated with recommendations centred around future milestones, particularly city construction, settlement expansion and resource accumulation. Participants appeared to adopt several of these planning concepts through repeated exposure to tutor recommendations. This suggests that milestone-based planning can effectively communicate long-term strategic goals by linking individual actions to future objectives.

Explaining Relative Preferences Is More Difficult While participants generally accepted the tutor's recommendations, several criticisms focused on understanding why the selected action was preferable to a reasonable alternative. This highlights a distinction between communicating a strategic objective and explaining the relative preference between competing actions. The underlying evaluation function considers multiple heuristics simultaneously, whereas explanations must compress this reasoning into concise natural language.

Richer Explanations Introduce a New Trade-Off A natural response to these limitations would be to generate increasingly detailed explanations. However, richer explanations often require additional abstraction, increasing the risk that explanations become generic or only loosely connected to the underlying decision process.

For example, an explanation might state that building a city improves resource income, supports expansion and creates stronger future opportunities. While this appears informative, a new question emerges: was the city actually selected for *all* of those reasons?

This introduces a trade-off between faithfulness, specificity and readability. As explanations become richer, they may increasingly resemble plausible strategic narratives rather than faithful accounts of the factors that influenced the decision.

Convenience Versus Reflection Participants emphasised the value of immediately seeing the recommended move, yet the clearest changes in reasoning occurred when players first formed their own decision and then compared it against the tutor's recommendation.

This suggests that features which maximise convenience may not necessarily maximise educational value. A performance-focused system would reveal recommendations immediately, whereas an education-focused system may benefit from encouraging reflection before revealing the preferred action.

4.7 Threats to Validity

Several limitations affect the extent to which the results can be generalised.

Internal Validity *Catan* is inherently stochastic due to dice rolls, hidden information and multi-agent interaction. Although fixed random seeds and repeated simulations were used to reduce variance, gameplay outcomes remain sensitive to chance events and opponent behaviour.

The evaluation is also dependent on the assumptions embedded within the rule-based framework. Tutor recommendations, move-quality scores and explanations are all derived from the same underlying heuristic evaluation function. As a result, alternative but reasonable play styles may sometimes be judged unfavourably.

External Validity Most gameplay evaluation was conducted using simulated agents rather than human opponents. While this enables controlled experimentation, simulated policies cannot fully capture the negotiation, unpredictability and social dynamics present in human gameplay.

The human-centred evaluation was also limited in scale, involving five participants who were primarily beginner or casual players. Consequently, the findings should be interpreted as exploratory rather than representative of the wider *Catan* player population.

Construct Validity Gameplay metrics such as win rate and average victory points provide useful measures of performance but do not directly capture explanation quality or educational effectiveness. Similarly, move-quality scores measure consistency with the tutor's heuristic framework rather than objective correctness.

Furthermore, many situations in *Catan* admit multiple strategically reasonable actions. This makes it difficult to determine whether a move is genuinely suboptimal or simply reflects a different strategic preference.

Chapter 5

Discussion and Conclusions

5.1 Discussion

The evaluation suggests that the project successfully achieved its primary technical objectives. The ETW-based decision framework consistently outperformed simpler baseline agents, while CMA-ES optimisation further improved gameplay performance (Table 4.1). The resulting simulation environment also provided a flexible platform for experimentation, supporting large-scale tournaments, ablation studies and decision-level analysis.

The project additionally demonstrated that rule-based planning can support explainable tutoring in a complex multiplayer game. Because recommendations were generated from interpretable strategic components, the tutor was able to expose reasoning in terms of resource production, expansion opportunities, milestone progression and opponent interference. This provided users with explanations grounded directly in the same framework used for decision-making rather than relying on post-hoc interpretation.

One of the most unexpected findings concerned how participants appeared to gain strategic understanding from the tutor. Initially, explanations were expected to be the primary educational mechanism. However, qualitative feedback suggested that users often developed strategic insight by comparing their intended actions against the tutor's recommendations (Section 4.6.2). Changes in resource management, city prioritisation and opening settlement placement were frequently associated with repeated exposure to alternative decisions rather than explanation content alone.

A similar pattern emerged regarding trust. Participants generally understood that *Catan* contains substantial randomness and that following good advice does not guarantee victory. Nevertheless, confidence in the tutor appeared to increase when recommendations consistently produced stronger positions, improved openings and faster progression towards victory points (Section 4.6.3). This suggests that perceived competence may play an important role in establishing trust within strategic tutoring systems.

The evaluation also highlighted a limitation of the explanation approach. Participants often accepted the tutor's overall strategic objective but still wanted additional justification for why one reasonable action was preferred over another. Explaining strategic goals therefore proved easier than explaining relative preferences between alternatives. This may represent an important challenge for future tutoring systems, particularly in domains where multiple viable strategies coexist.

Beyond the tutoring aspects, the results reinforced the importance of opening settlement placement within *Catan*. Performance differences between agents narrowed substantially when weaker agents were provided with stronger opening positions (Table 3.1), suggesting that early-game decisions exert a disproportionate influence on later strategic opportunities. This was also one of the areas where participants reported the greatest benefit from tutor assistance.

More broadly, the project highlighted the difficulty of evaluating decision quality in stochastic multiplayer games. Compared to deterministic environments, move quality is more strongly influenced by uncertainty, hidden information and opponent behaviour. As a result, multiple strategically reasonable actions may coexist within the same position. Consequently, tutor recommendations should be viewed as model-based guidance rather than objective judgements of correctness.

5.2 Limitations

Despite the positive results, the project has several important limitations.

First, the rule-based framework reflects assumptions embedded by the designer regarding what constitutes an effective strategy. Although evolutionary optimisation improves the balance between heuristics, the overall strategic space explored by the agent remains constrained by the manually designed evaluation structure. As a result, the tutor may fail to generalise to alternative play styles or unconventional but effective strategies.

Second, the ETW and ETB approximations simplify several important aspects of real gameplay. Negotiation behaviour, hidden information and long-term opponent adaptation are only partially modelled within the current framework. While these approximations improve computational tractability and support interpretability, they limit the realism of strategic evaluation in highly interactive game states.

The deterministic nature of the decision-making process also introduces limitations. Agents with identical weights and heuristics tend to produce similar strategic preferences, which can reduce behavioural diversity across repeated games. While optional stochastic action selection can be used at lower difficulty levels to introduce variation, the underlying policy remains largely deterministic and may allow experienced players to anticipate recurring strategic patterns.

Although explanations are faithful to the utility decomposition used by the agent, the evaluation suggested that users often wanted deeper justification between competing actions than the current explanation framework could provide. Explanations currently focus on communicating the tutor's reasoning process rather than adapting to the learner's level of expertise, misconceptions or preferred learning style.

Evaluation was additionally limited in scope. Human-centred evaluation primarily measured perceived usefulness and interpretability during short-term interaction with the tutor. The project did not measure long-term changes in player skill or strategic understanding, nor compare the tutor directly against human coaching. A more robust educational evaluation would require longer-term studies measuring how player strategy and decision-making change through repeated use of the tutor. The participant sample size was also relatively limited, reducing the extent to which conclusions can be generalised.

Finally, the project focused primarily on strategic decision-making outside of complex social interaction. While trading and opponent interference are partially modelled, the tutor does not fully capture the psychological and negotiation-driven aspects of human *Catan* play, which are often central to high-level gameplay.

5.3 Future Work

Several directions for future work emerge from the limitations of the current system.

One natural extension would be to incorporate adaptive tutoring behaviour. Rather than evaluating moves solely according to a fixed utility framework, future systems could model player intentions, skill level and historical behaviour in order to provide more personalised feedback. This could allow explanations to become more conversational and educationally targeted.

Hybrid architectures combining symbolic reasoning with learning-based methods also represent a promising direction. For example, reinforcement learning or search-based systems could be used to improve strategic strength while retaining interpretable high-level reasoning components for explanation generation.

The explanation system itself could additionally be extended beyond static feedback. Interactive dialogue-based explanations may allow players to question recommendations, compare alternative strategies or request clarification about long-term plans. The tutor study suggested that users often wanted to understand broader strategic concepts beyond a single recommendation, such as development card timing and long-term strategic priorities. Such interaction may better approximate the behaviour of a human tutor and provide greater educational value than static explanations alone.

Future work could also improve opponent modelling and negotiation reasoning. More sophisticated representations of trading behaviour, bluffing and social dynamics may produce more realistic gameplay and improve the tutor's ability to evaluate strategically ambiguous situations.

Finally, larger-scale user studies would help evaluate the educational effectiveness of the tutor more rigorously. In particular, future evaluation could measure long-term player improvement, strategic understanding and trust in AI-generated explanations across players of different skill levels.

5.4 Conclusion

This project developed a complete explainable AI tutor for the board game *Catan*, integrating gameplay simulation, ETW-driven strategic planning, evolutionary optimisation, explanation generation and interactive tutoring within a unified framework. The resulting system demonstrated that interpretable rule-based planning can produce strategically competent gameplay while simultaneously exposing the reasoning behind recommendations in a form accessible to human players.

Beyond the technical implementation, the project provided several insights into the design of explainable tutoring systems for complex strategy games.

Firstly, participants appeared to develop strategic understanding primarily through comparison with tutor recommendations rather than explanations alone, suggesting that exposure to stronger alternatives may be a key mechanism through which tutors influence player decision-making.

Secondly, trust appeared to be influenced more by perceived competence than explanation quality. Participants generally became more confident in the tutor when its recommendations consistently produced stronger positions and meaningful strategic progress.

Thirdly, generating explanations proved easier than generating satisfying explanations. While strategic objectives could often be communicated effectively, explaining why one reasonable action should be preferred over another remained a significant challenge.

Finally, the project highlighted that decision quality in stochastic multiplayer games is inherently model-dependent, meaning tutor recommendations are best interpreted as guidance rather than objective judgements of correctness.

Overall, the project demonstrates that explainable game AI can provide useful strategic guidance while also exposing important questions surrounding trust, explanation design and decision evaluation. The resulting system provides a foundation for future work exploring adaptive tutoring, interactive explanation systems and AI-assisted strategy learning in stochastic multiplayer environments.

Declarations

Use of Generative AI

I acknowledge the use of ChatGPT during the development of this project. ChatGPT was used for feedback on report structure and clarity, discussion of technical concepts and occasional debugging assistance during implementation.

Generative AI tools were also used to assist in the creation of certain visual assets used within the game interface, including board-game-style graphics and interface imagery.

All writing, system design, implementation, experiments and analysis presented in this report are my own work. Generative AI tools were used only as a supporting aid and not to generate substantial portions of the report or project code without review and modification.

Ethical Considerations

This project involved the development of an explainable AI tutoring system for the board game *Catan*. The project did not involve sensitive personal data or high-risk human subjects research.

Where participants were involved in the evaluation of the tutoring system, participation was voluntary and no personally identifying information was collected or stored.

Sustainability

This project used a lightweight explainable AI approach based on weighted rules and evolutionary optimisation rather than large-scale deep learning models, reducing computational and energy requirements.

The tutor system runs locally as a desktop application and does not require continuous cloud-based computation during normal use.

Availability of Code and Materials

The project source code, runnable tutor system and evaluation scripts are available at:

https://github.com/AlexShem247/a-catan-tutor

Bibliography

[1] Papadam DR, Chalkiadakis G. Adversarial search and deep learning for strategic settlement placement in Settlers of Catan. In: *Proceedings of the European Conference on Multi-Agent Systems (EUMAS 2024)*. Springer; 2025. p. 57-76.

[2] Szita I, Chaslot G, Spronck P. Monte Carlo tree search in Settlers of Catan. In: *Advances in Computer Games*. Springer; 2010. p. 21-32.

[3] Dobre MS. Low-resource learning in complex games. University of Edinburgh. PhD thesis; 2018.

[4] Gendre Q, Kaneko T. Playing Catan with cross-dimensional neural network. In: *Neural Information Processing*. Springer; 2020. p. 580-92. Available from: https://arxiv.org/abs/2008.07079.

[5] Daijavad S, Kaplan S, Kelleran J, Smith A, Vu T. From resources to victory: heuristic-driven reinforcement learning in Catan. California Polytechnic State University. Student project report; 2025.

[6] Guhe M, Lascarides A. Game strategies for the Settlers of Catan. In: *Proceedings of the 2014 IEEE Conference on Computational Intelligence and Games*. IEEE; 2014. p. 1-8.

[7] Catan Studio. Catan game rules & almanac; 2020. Accessed 2025-11-23. Available from: https://www.catan.com/sites/default/files/2021-06/catan_base_rules_2020_200707.pdf.

[8] Thomas R. Real-time decision making for adversarial environments using a plan-based heuristic. Northwestern University. PhD thesis; 2003.

[9] Osband I, Van Roy B, Wen Z. Generalization and exploration via randomized value functions; 2016. Available from: https://arxiv.org/abs/1602.04621.

[10] Pfeiffer M. Machine learning applications in computer games. *ÖGAI-Journal*. 2004;23(3):4-7.

[11] Austin J, Kronenthal BG, Miller SM. Off to a good start: using mathematics to evaluate settlement locations in Catan. In: Capaldi M, editor. *Teaching mathematics through games*. American Mathematical Society; 2021. p. 137-43.

[12] Xu Y. Dialogue explanations for rule-based AI systems. In: *Explainable and transparent AI and multi-agent systems*. vol. 14127 of Lecture Notes in Computer Science. Springer Nature; 2023. p. 59-77.

[13] Juozapaitis Z, Koul A, Fern A. Explainable reinforcement learning via reward decomposition. In: *Proceedings of the IJCAI/ECAI Workshop on Explainable Artificial Intelligence*; 2019. p. 47-53.

[14] Chess.com Support. How does Game Review work?; 2025. Accessed 2025-11-23. Available from: https://support.chess.com/en/articles/8584089-how-does-game-review-work.

[15] Hammersborg P, Strumke I. Information based explanation methods for deep learning agents: with applications on large open-source chess models. *Scientific Reports*. 2024;14(1):20174. Available from: https://arxiv.org/abs/2309.09702.

[16] Putnam V, Riegel L, Conati C. Toward XAI for intelligent tutoring systems: a case study; 2019. Available from: https://arxiv.org/abs/1912.04464.

[17] Karpouzis K. Explainable AI for intelligent tutoring systems. In: *International Conference on Frontiers of Artificial Intelligence, Ethics, and Multidisciplinary Applications*. Springer; 2023. p. 59-70.

[18] Min W, Mott B, Lester J. Adaptive scaffolding in an intelligent game-based learning environment for computer science. In: *Proceedings of the Workshop on AI-Supported Education for Computer Science (AIEDCS)*; 2014. p. 41-50.

[19] Aleven V, McLaughlin EA, Glenn RA, Koedinger KR. Instruction based on adaptive learning technologies. In: *Handbook of research on learning and instruction*. Routledge; 2016. p. 477-518.

[20] Khosravi H, Buckingham Shum S, Chen G, et al. Explainable artificial intelligence in education. *Computers and Education: Artificial Intelligence*. 2022;3:100074.

[21] Mueller ST, Hoffman RR, Clancey W. Explanation in human-AI systems: a literature meta-review, synopsis of key ideas and publications, and bibliography for explainable AI; 2019. Available from: https://arxiv.org/abs/1902.01876.

[22] Rong Y, Leemann T, Nguyen TT, et al. Towards human-centered explainable AI: a survey of user studies for model explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2024;46(4):2104-22. Available from: https://arxiv.org/abs/2210.11584.

[23] Maddison CJ, Huang A, Sutskever I, Silver D. Move evaluation in Go using deep convolutional neural networks; 2014. Available from: https://arxiv.org/abs/1412.6564.

[24] Thaker CS, Jat DS, Xoagub AJ. Evolutionary computation in artificial board game playing through genetic weight evolution. In: *Proceedings of the International Conference on Emerging Trends in Networks and Computer Communications (ETNCC)*. IEEE; 2015. p. 62-6.

[25] Risi S, Togelius J. Neuroevolution in games: state of the art and open challenges. *IEEE Transactions on Computational Intelligence and AI in Games*. 2017;9(1):25-41.

[26] Hansen N, Ostermeier A. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*. 2001;9(2):159-95.

[27] Gamma E, Helm R, Johnson R, Vlissides J. *Design patterns: elements of reusable object-oriented software*. Addison-Wesley; 1994.

[28] Patel A. Hexagonal grids; 2013. Accessed 2026-02-04. Available from: https://www.redblobgames.com/grids/hexagons/.

[29] Riverbank Computing. PyQt6: Python bindings for the Qt 6 application framework; 2021. Accessed 2025-11-17. Available from: https://www.riverbankcomputing.com/software/pyqt/.

[30] dos Santos Mignon A, de Azevedo da Rocha RL. An adaptive implementation of ϵ-greedy in reinforcement learning. *Procedia Computer Science*. 2017;109:1146-51.